The Thiele Machine

Exact Statements

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Abstract

The book, *The Thiele Machine: No Free Insight*, argues everything in plain words, with the reader, and leaves out theorem statements on purpose. This companion gives the exact statements: the definitions with every quantifier written out, and each result the book shows, with its hypotheses, in the book's order. Each result is tagged with its kind from the ledger of Chapter 22: [shown] for an argument, [chosen] for a definition or a price, [assumed] for a claim about the world that an argument leans on, [cited] for somebody else's result, and [thesis] for a claim no argument can settle. Where a proof is short it is printed; otherwise the pointer says where the book argues it. Nothing here goes beyond the book. A statement that looks stronger than the book's is a mistake in this document.

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1 Notation

$\mathbb{N} = \{0, 1, 2, \dots\}$. For a finite set X, $|X|$ is its size. $\log_2$ is the logarithm base two and ln the natural logarithm.

A *machine* is a triple (S, I, δ): a set S of states, a set I of moves, and a step $\delta : S \times I \to S$. A machine with *one next state* is the case $|I| = 1$, written as a function $f : S \to S$. A *run* from s along a list of moves $\tau = i_1 \cdots i_n$ is the list of states $s_0 = s$, $s_k = \delta(s_{k-1}, i_k)$; $\mathrm{run}(\tau, s) = s_n$.

Two arrows *meet* at a move i when there are $s \neq s'$ with $\delta(s, i) = \delta(s', i)$; the usual word is a *merge*. For $f : S \to S$, two arrows meet when $f(s) = f(s')$ for some $s \neq s'$, that is, when f is not one-to-one.

A *reading* is a function $r : S \to R$ into a preorder $(R, \leq)$: a reflexive, transitive relation. The *two-point* reading takes values in $\{\mathrm{no}, \mathrm{yes}\}$ with $\mathrm{no} < \mathrm{yes}$. A move i *settles more* at s when $r(\delta(s, i)) \not\leq r(s)$. On the two-point reading that is exactly $r(s) = \mathrm{no}$ and $r(\delta(s, i)) = \mathrm{yes}$.

A *price list* is $c : I \to \mathbb{N}$. Where a result lets the price depend on the state a step starts from, it says so and writes $c : S \times I \to \mathbb{N}$. The price of a run is the sum of the prices of its moves.

A *window* is a function $\pi : S \to B$. The *pile* behind a picture $b \in B$ is $\pi^{-1}(b)$.

2 The definition

Definition 2.1 (The toll)**.** [chosen] A price list c *obeys the toll* for a reading r on a machine (S, I, δ) when

$$\forall s \in S,\ \forall i \in I: \quad r(\delta(s,i)) \not\leq r(s) \implies c(i) \geq 1.$$

Definition 2.2 (Thiele machine)**.** [chosen] A *Thiele machine* is a machine (S, I, δ) with a reading $r : S \to (R, \leq)$, a price list $c : I \to \mathbb{N}$, and a set $S_0 \subseteq S$ of starts, such that

(i) (settled never goes back) $r(s) \leq r(\delta(s,i))$ for every $s \in S$ and $i \in I$;

(ii) (something settles) there are $s_0 \in S_0$ and a list of moves τ with $r(\mathrm{run}(\tau, s_0)) \not\leq r(s_0)$;

(iii) (the toll) c obeys the toll for r.

Prices are in $\mathbb{N}$: nothing or more, never a refund.

Remark 2.3. Clause (ii) rules out a machine that never settles anything, which would meet (i) and (iii) for free. Prices in $\mathbb{N}$ rule out a later move paying back an earlier charge. The smallest reading is the two-point one, and most results below are stated on it; a checklist of k boxes whose ticks never come off is the reading into $\{\mathrm{no}, \mathrm{yes}\}^k$ ordered coordinatewise.

Argued in Chapter of the book.

Definition 2.4 (Established record)**.** [chosen] Let a machine carry a two-point reading r (the box), a set of *claims* C, for each claim $\kappa \in C$ a set of check moves and a relation saying which moves *touch* what κ is about, and a predicate saying whether a check of κ *passes* at a state. A start is *clean* when its box is down. The box is *established for* κ when, on every run from a clean start, every step that turns the box up is preceded in that run by a step that is a passing check of κ, with no move that touches what κ is about between that check and the raise. The box is still a reading of a Thiele machine, so the step that turns it up pays the toll. That the check means what it says and could have failed is a fact about the checker, not a condition on the machine.

Argued in Chapter 8 of the book.

3 Any machine that steps

Throughout this section $f : S \to S$ is a machine with one next state and $x_0 \in S$ is a start. The run from x_0 is $x_n = f^n(x_0)$ for $n \in \mathbb{N}$.

Definition 3.1 (Fresh steps, meeting on a run)**.** [chosen] The step from x_n is *fresh* when $x_{n+1} \notin \{x_0, \ldots, x_n\}$. Two arrows *meet on the run within* n when there are $i, j \leq n$ with $x_i \neq x_j$ and $f(x_i) = f(x_j)$. Two arrows meet on the run when they meet on it within some n.

Definition 3.2 (Turing's machine)**.** [cited] Turing's model, in the form used here: a finite alphabet Σ with a blank $\sqcup \in \Sigma$, a finite set Q of letters, and a *table*, a partial function $\tau : Q \times \Sigma \rightharpoonup Q \times \Sigma \times \{-1, 0, +1\}$. A *configuration* is a triple (q, h, w) with $q \in Q$, $h \in \mathbb{Z}$ the head, and $w : \mathbb{Z} \to \Sigma$ the tape, blank at all but finitely many cells. If $\tau(q, w(h)) = (q', \sigma, d)$, the next configuration is $(q', h + d, w')$, where $w'(h) = \sigma$ and $w'(p) = w(p)$ for $p \neq h$. If $\tau(q, w(h))$ is undefined, the configuration has no next configuration and is *stopped*.

Remark 3.3. A configuration is the whole state: the table does not change, and the next configuration is a function of the configuration alone. So a Turing machine is a machine with at most one next state, with the configurations as states. The claims that other models of computation compute the same functions as Turing's are [cited]. The Church-Turing thesis, and the claim that every machine that can be built is covered, are claims about the world; no result in this document rests on them.

Argued in Chapter 1 of the book.

Proposition 3.4 (The step law)**.** [shown] *Let $f : S \to S$ and $x_0 \in S$, with no condition on S. For every $n \in \mathbb{N}$ the step from x_n does one of the following, and (a) excludes (b) and (c):*

(a) it is fresh;

(b) $x_{n+1} = x_j$ for some $j \leq n$ with $j = 0$ or $x_{j-1} = x_n$ (it goes round a loop: back to the start, or along a step the run has taken before);

(c) $x_{n+1} = x_j$ for some $1 \leq j \leq n$ with $x_{j-1} \neq x_n$ (two arrows meet: $f(x_{j-1}) = f(x_n)$).

If some step is not fresh, let the step from x_n be the first, landing on x_j with $j \leq n$. Every earlier step is fresh; either $j = 0$, or $j \geq 1$ and $x_{j-1} \neq x_n$, so two arrows meet; and from x_j on the run repeats with period $p = n + 1 - j$: $x_{k+p} = x_k$ for all $k \geq j$.

Proof. If $x_{n+1} \notin \{x_0, \ldots, x_n\}$, (a) holds. Otherwise $x_{n+1} = x_j$ for some $j \leq n$; if $j = 0$ or $x_{j-1} = x_n$ that is (b), and else (c). At the first revisit, $x_0, \ldots, x_n$ are pairwise distinct, so for $j \geq 1$ the states x_{j-1} and x_n are distinct ($j - 1 < n$), and both step to x_j. Since f is a function, $x_{n+1} = x_j$ gives $x_{n+1+m} = x_{j+m}$ for every m, which is the period. □

Argued in Chapter 1 of the book.

Proposition 3.5 (Keeping on any number of states)**.** [shown] *Let $f : S \to S$, and let $r : S \to \{\text{no}, \text{yes}\}$ be a two-point reading that never goes back under f: $r(s) = \text{yes}$ implies $r(f(s)) = \text{yes}$. Let $r(x_0) = \text{no}$ and $r(x_m) = \text{yes}$ for some m. Then $x_k \neq x_0$ for every $k \geq 1$, and either the states $x_0, x_1, x_2, \ldots$ are pairwise distinct, or two arrows meet on the run. No condition on S is used.*

Proof. If $x_k = x_0$ with $k \geq 1$, then $x_{qk} = x_0$ for every $q \in \mathbb{N}$. Take q with $qk \geq m$; since the reading never goes back, $r(x_{qk}) = \text{yes} \neq r(x_0)$, a contradiction. So no step goes round a loop through the start. If the states are not pairwise distinct, some step is not fresh, and by Proposition 3.4 the first such step lands on x_j with $j \geq 1$ and makes two arrows meet. □

Argued in Chapter 1 of the book.

Proposition 3.6 (A machine that stops slips past the merge)**.** [shown] *Let $S = \{w, h\}$ with $w \neq h$, a partial step f with $f(w) = h$ and no next state for h, and $r(w) = \text{no}$, $r(h) = \text{yes}$. The run from w raises the reading, keeps it, visits two states, and no two arrows meet. Let $g : S' \to S'$ be any total step on a set $S' \supseteq S$ with $g(w) = h$, and r extended to a two-point reading on S' that never goes back under g. Then the run of g from w visits pairwise distinct states forever, or two arrows meet on it. In particular, $g(h) = h$ gives $g(w) = g(h)$; and $g(h) = p$, $g(p) = p$ with $p \notin S$ gives $g(h) = g(p)$.*

Proof. The partial step has one arrow, so no two arrows meet. The statement about g is Proposition 3.5 for g from w. The two particular cases are direct. □

Argued in Chapter 1 of the book.

Proposition 3.7 (A machine that stops, with a price on stopping, is a Thiele machine)**.** [shown] *Let $f : S \rightharpoonup S$ be a partial step and $H = S \setminus \mathrm{dom}(f)$ its stopped states. Let $\bar{f}(s) = f(s)$ for $s \notin H$ and $\bar{f}(s) = s$ for $s \in H$ (a stopped state stays where it is). Read it through whether it has stopped: $r(s) =$ yes if and only if $s \in H$. Let $S_0 \subseteq S$, and suppose some $s_0 \in S_0 \setminus H$ reaches H, that is, $\bar{f}^{\,n}(s_0) \in H$ for some n. Then:*

(a) with one move t and $c(t) = 1$, the tuple $(S, \{t\}, \bar{f}, r, c, S_0)$ is a Thiele machine, and every price list that obeys the toll for r has $c(t) \geq 1$;

(b) every step that stops it, from $s \notin H$ to $f(s) \in H$, is a step where two arrows meet: $s \neq f(s)$ and $\bar{f}(s) = \bar{f}(f(s))$.

This applies to a Turing machine, with its configurations as states, and to any machine with at most one next state. The only datum added to the machine is the price.

Proof. (a) If $r(s) =$ yes then $s \in H$ and $\bar{f}(s) = s \in H$, so clause (i) holds. The run of length n from s_0 ends in H while $s_0 \notin H$, so clause (ii) holds. Clause (iii) holds since $c(t) = 1$. Some step of that run goes from no to yes, and it is a t-step, so the toll forces $c(t) \geq 1$. (b) $s \notin H$ and $f(s) \in H$ give $s \neq f(s)$, and $\bar{f}(f(s)) = f(s) = \bar{f}(s)$. □

Argued in Chapter of the book.

Argued in Chapter 1 of the book.

Proposition 3.8 (Turing's model makes no mark hold)**.** [shown] *In Turing's model (Definition 3.2), let τ be a table, (q, h, w) a configuration, and $\sigma \in \Sigma$ any symbol. The table τ' that agrees with τ everywhere except at the entry $(q, w(h))$, where $\tau'(q, w(h)) = (q, \sigma, 0)$, is a table of the model, and under τ' the next configuration carries σ at cell h. So for every cell and every mark, the model contains a table that rubs the mark out ($\sigma = \sqcup$) and a table that prints it, whatever came before. No condition of the model makes a mark, once written, stay.*

Proof. Every partial function $Q \times \Sigma \rightharpoonup Q \times \Sigma \times \{-1, 0, +1\}$ is a table, so τ' is one, and the step rule of Definition 3.2 writes σ at h. □

Argued in Chapter of the book.

Argued in Chapter 22 of the book.

4 Running out of room

For a two-point reading r write $Y = r^{-1}(\text{yes})$, the *done* states. For a move i write $\delta_i = \delta(\cdot, i)$. A move i is *one-to-one* when δ_i is; otherwise two arrows meet at i. A move i *keeps done states done* when $\delta_i(Y) \subseteq Y$. Clause (i) of Definition 2.2, on a two-point reading, says that every move keeps done states done.

Proposition 4.1 (The chairs)**.** [shown] *Let (S, I, δ) be a machine with a two-point reading r, and Y finite. Let i be a move that keeps done states done, and $s \notin Y$ with $\delta(s, i) \in Y$. Then two arrows meet at i: there are $t \neq t'$ in $Y \cup \{s\}$ with $\delta(t, i) = \delta(t', i)$.*

Proof. δ_i maps the $|Y|+1$ states of $Y \cup \{s\}$ into Y, which has $|Y|$ states. By the pigeonhole principle two of them have the same image. □

Argued in Chapter 2 of the book.

Example 4.2 (The meeting need not be at the landing)**.** $S = \{3, 4, 5a, 5b\}$, $f(3) = 4$, $f(4) = 5a$, $f(5a) = 5b$, $f(5b) = 5b$, $Y = \{5a, 5b\}$. The step from 4 settles the answer, and no other state lands on $5a$. Two arrows meet at $5b$, from $5a$ and $5b$, one tick later on the run. The move as a whole is not one-to-one, as Proposition 4.1 requires.

Argued in Chapter of the book.

Proposition 4.3 (Merge or fresh states, counted)**.** [shown] *Let $f : S \to S$, r a two-point reading that never goes back under f, $r(x_0) =$ no, and $r(x_m) =$ yes for some m. If two arrows do not meet on the run within n, then $x_0, \ldots, x_{n+1}$ are pairwise distinct. Consequently, if S is finite, two arrows meet on the run within $|S| - 1$.*

Proof. Suppose $x_i = x_j$ with $i < j \leq n+1$, and take such a pair with i least. If $i \geq 1$, then $f(x_{i-1}) = f(x_{j-1})$ with $i-1, j-1 \leq n$, so $x_{i-1} = x_{j-1}$ since no two arrows meet within n; this contradicts the choice of i. So $x_j = x_0$ with $j \geq 1$, which Proposition 3.5 excludes. For the consequence, take $n = |S| - 1$: the $|S|+1$ states $x_0, \ldots, x_{|S|}$ cannot be pairwise distinct. □

Argued in Chapter 2 of the book.

Proposition 4.4 (Endless room: Hilbert's hotel)**.** [shown]

(a) On $S = \mathbb{N}$ let $f(n) = n+1$ and $r(n) =$ yes if and only if $n \geq 1$ (the strip: n marks). Then f is one-to-one, the reading never goes back, the step from 0 raises it, and every step of every run is fresh.

(b) Conversely, let $f : S \to S$ be one-to-one, r a two-point reading that never goes back under f, and $r(s) =$ no, $r(f(s)) =$ yes. Then the states $f^t(s)$, $t \in \mathbb{N}$, are pairwise distinct, and f maps Y one-to-one into $Y \setminus \{f(s)\}$. So Y is infinite.

Proof. (a) is direct. (b) The iterates are distinct by Proposition 3.5, since a one-to-one f never makes two arrows meet. $f(Y) \subseteq Y$ because the reading never goes back, and $f(s) \notin f(Y)$ because f is one-to-one and $s \notin Y$. A finite set has no one-to-one map onto a proper subset of itself. □

Argued in Chapter 2 of the book.

Proposition 4.5 (Nothing meets on finitely many states: back to the start)**.** [shown] *Let S be finite and $f : S \to S$ one-to-one. Every run returns to its start: $x_p = x_0$ for some $1 \leq p \leq |S|$. Consequently, if a two-point reading never goes back under f and $r(x_0) =$ no, then $r(x_k) =$ no for every k: the machine keeps nothing.*

Proof. Among $x_0, \ldots, x_{|S|}$ two are equal, $x_i = x_j$ with $i < j \leq |S|$. Cancelling f i times, which is allowed because f is one-to-one, gives $x_0 = x_{j-i}$. If $r(x_k) =$ yes for some k, then the reading is yes at every multiple of $p = j - i$ that is at least k, where the run is at x_0, a contradiction. □

Argued in Chapter 2 of the book.

Corollary 4.6 (Several moves, none meeting)**.** [shown] *Let (S, I, δ) have S finite and every move one-to-one, and let a reading r into a preorder satisfy clause (i) of Definition 2.2. Then $r(\mathrm{run}(\tau, s)) \le r(s)$ for every s and every list τ. So clause (ii) fails, and the machine is not a Thiele machine for any price list and any starts.*

Proof. Each δ_i is a permutation of the finite set S, so $\delta_i^{p_i}$ is the identity for some $p_i \ge 1$, and $\delta_i^{-1} = \delta_i^{p_i - 1}$. Hence for $t = \mathrm{run}(\tau, s)$ there is a list τ' with $\mathrm{run}(\tau', t) = s$. Clause (i) along τ' gives $r(t) \le r(s)$. □

Argued in Chapter 2 of the book.

Example 4.7 (The cut strip)**.** Let $S = \{0, 1, \ldots, 8\}$ (the number of marks on an eight-box strip), $f(n) = n + 1$ for $n < 8$, and $r(n) =$ yes if and only if $n \ge 1$. Leaving $f(8)$ undefined is a machine that stops. Every choice of $f(8) \in S$ either lowers the reading ($f(8) = 0$) or makes two arrows meet: if $f(8) = k \ge 1$ then $f(k - 1) = k = f(8)$ and $k - 1 \ne 8$.

Argued in Chapter 2 of the book.

Proposition 4.8 (How long a finite machine can hold off)**.** [shown] *Let $f : S \to S$, r a two-point reading that never goes back under f, Y finite with $|Y| = K$, $x_0 \notin Y$ and $x_1 \in Y$. Then two arrows meet on the run within K. The bound is exact: for every $K \ge 1$, the machine $S = \{u, h_1, \ldots, h_K\}$ with $f(u) = h_1$, $f(h_j) = h_{j+1}$ for $j < K$, $f(h_K) = h_K$, $Y = \{h_1, \ldots, h_K\}$, run from u, has no two arrows meeting within $K - 1$, and two arrows meeting within K.*

Proof. If two arrows do not meet within K, Proposition 4.3 makes $x_1, \ldots, x_{K+1}$ pairwise distinct, and all lie in Y because the reading never goes back: $K + 1$ states in a set of K. In the example, $x_0, \ldots, x_{K-1}$ are $u, h_1, \ldots, h_{K-1}$, with the distinct images $h_1, \ldots, h_K$, and $f(h_{K-1}) = f(h_K)$ (for $K = 1$, $f(u) = f(h_1)$). □

Argued in Chapter 2 of the book.

Remark 4.9. Whether a move is one-to-one is a fact about the move on a stated set of states. Every merge in this document is counted on the whole state set the machine is given, and on nothing outside it.

5 Why charge for it

The notation of Section 4 is kept: $Y = r^{-1}(\text{yes})$ and $\delta_i = \delta(\cdot, i)$.

Definition 5.1 (Bennett's rule)**.** [chosen] A price list $c : I \to \mathbb{N}$ *follows Bennett's rule* on (S, I, δ) when every move at which two arrows meet costs at least one:

$$\forall i \in I : \quad \delta_i \text{ not one-to-one} \implies c(i) \ge 1.$$

The rule is Bennett's: what has to cost is the merging of two computation paths.

Argued in Chapter 3 of the book.

Theorem 5.2 (The toll follows)**.** [shown] *Let (S, I, δ) carry a two-point reading r with Y finite, satisfying clause (i) of Definition 2.2 (done stays done), and let c follow Bennett's rule. Then c obeys the toll for r.*

Proof. If $r(s) = \text{no}$ and $r(\delta(s,i)) = \text{yes}$, Proposition 4.1 applies, since every move keeps done states done. So two arrows meet at i, and Bennett's rule gives $c(i) \geq 1$. □

Argued in Chapter 3 of the book.

Proposition 5.3 (Merge or take back)**.** [shown] *Let (S, I, δ) carry a two-point reading r with Y finite. No condition that done stays done is assumed. If $r(s) = \text{no}$ and $r(\delta(s,i)) = \text{yes}$, then two arrows meet at i, or there is $t \in Y$ with $\delta(t,i) \notin Y$.*

Proof. If no such t exists, i keeps done states done, and Proposition 4.1 applies. □

Argued in Chapter 3 of the book.

Example 5.4 (Charged is not the same as kept)**.** $S = \{T_0, T_1, T_2\}$, $Y = \{T_1\}$, one move f with $f(T_0) = T_1$, $f(T_1) = T_2$, $f(T_2) = T_1$. The move settles at T_0, takes back at T_1, and makes two arrows meet ($f(T_0) = f(T_2)$). Bennett's rule charges it, and it keeps nothing. The implication of Theorem 5.2 has no converse.

Argued in Chapter 3 of the book.

Proposition 5.5 (The moves Bennett's rule forces are the merges)**.** [shown] *Let (S, I, δ) be any machine, finite or not, and $i \in I$. Then $c(i) \geq 1$ for every price list c that follows Bennett's rule if and only if two arrows meet at i.*

Proof. If two arrows meet at i, the rule itself charges i. If δ_i is one-to-one, the price list with $c(i) = 0$ and $c(j) = 1$ for $j \neq i$ follows the rule and does not charge i. □

Argued in Chapter 3 of the book.

Corollary 5.6. [shown] *Under the hypotheses of Theorem 5.2, every move that settles more somewhere is charged by every price list that follows Bennett's rule.*

Argued in Chapter 3 of the book.

Proposition 5.7 (Three doors)**.** [shown] *Each of the three conditions of Theorem 5.2 (finitely many done states, done stays done, Bennett's rule) can fail while the other two hold and some move settles more, and then the toll can fail.*

(a) The strip: $S = \mathbb{N}$, one move $n \mapsto n+1$, $r(n) = \text{yes}$ if and only if $n \geq 1$, $c = 0$. Done stays done, Bennett's rule holds (the move is one-to-one), the step from 0 settles, the toll fails, and Y is infinite.

(b) The coin: $S = \{\text{tails}, \text{heads}\}$, one move that swaps them, $r(\text{heads}) = \text{yes}$, $c = 0$. Y is finite, Bennett's rule holds, the step from tails settles, the toll fails, and done does not stay done.

(c) The stamp at zero: $S = \{\text{blank}, \text{stamped}\}$, one move sending both to stamped, $r(\text{stamped}) = \text{yes}$, $c = 0$. Y is finite, done stays done, the step from blank settles, the toll fails, and Bennett's rule fails.

Proof. (a) $n + 1 = n' + 1$ gives $n = n'$; a state $n \geq 1$ goes to $n + 1 \geq 1$; the step from 0 settles at price 0; $Y = \{1, 2, \dots\}$. (b) The swap is a bijection; it sends heads to tails. (c) Both states land on stamped; the step from blank settles at price 0. □

Argued in Chapter 3 of the book.

Definition 5.8 (The halving rule)**.** [chosen] For a move i and $y \in S$, the *pile* of i at y is $\delta_i^{-1}(y)$. A price list c *follows the halving rule* when each unit of cost pays for at most one halving:

$$\forall i \in I,\ \forall D \subseteq S \text{ finite}: \quad |D| \le 2^{c(i)}\,|\delta_i(D)|.$$

Argued in Chapter 3 of the book.

Lemma 5.9 (The worst pile-up)**.** [shown] *c follows the halving rule if and only if, for every move i, every pile of i has at most $2^{c(i)}$ states. In particular a price list that follows the halving rule follows Bennett's rule, and a move with a pile of three states costs at least two.*

Proof. If every pile has at most $2^{c(i)}$ states, D is covered by its $|\delta_i(D)|$ piles, so $|D| \le 2^{c(i)}|\delta_i(D)|$. Conversely, a finite subset D of one pile has $|\delta_i(D)| = 1$, so $|D| \le 2^{c(i)}$; hence the pile is finite with at most $2^{c(i)}$ states. A pile of two forces $c(i) \ge 1$; a pile of three forces $2^{c(i)} \ge 3$. □

Argued in Chapter 3 of the book.

Proposition 5.10 (What a squeeze costs)**.** [shown] *Let r be a two-point reading with Y finite, $|Y| = m$, and i a move that keeps done states done. Let F be a set of $k \ge 1$ states outside Y with $\delta_i(F) \subseteq Y$, and let c follow the halving rule. Then*

$$m + k \le 2^{c(i)}\, m, \qquad \textit{that is,} \qquad c(i) \ge \log_2 \frac{m+k}{m},$$

and some state of Y is the image of at least $\lceil (m+k)/m \rceil$ states of $F \cup Y$. For $k = 1$ this gives $c(i) \ge 1$. The bound is attained: on $S = \{A, B, C, \mathrm{DONE}\}$ with $Y = \{\mathrm{DONE}\}$ and one move sending every state to DONE*, $m = 1$, $k = 3$, every price list following the halving rule has $c \ge 2$, and $c = 2$ follows it.*

Proof. $m \ge 1$ since $\delta_i(F) \subseteq Y$ and $F \neq \emptyset$. The $m+k$ states of $F \cup Y$ land in Y, so $|\delta_i(F \cup Y)| \le m$, and the halving rule gives $m + k \le 2^{c(i)} m$. The $m + k$ states are spread over at most m piles, so one pile has at least $(m + k)/m$ of them. In the example the single pile has four states, and Lemma 5.9 gives exactly $c \ge 2$. □

Argued in Chapter 3 of the book.

Proposition 5.11 (Counting a whole run)**.** [shown] *Let $D_0 \subseteq S$ be finite, $i_1, \ldots, i_n$ moves, and $D_t = \delta_{i_t}(D_{t-1})$. Then*

$$|D_0| = \sum_{t=1}^{n} (|D_{t-1}| - |D_t|) + |D_n|,$$

each term of the sum is at least 0, and if $|D_n| < |D_0|$ then some δ_{i_t} is not one-to-one on D_{t-1}: two arrows meet at i_t.

Proof. The sum telescopes. $|D_t| \le |D_{t-1}|$ because D_t is an image of D_{t-1}. If every δ_{i_t} is one-to-one on D_{t-1}, every term is 0 and $|D_n| = |D_0|$. □

Argued in Chapter 3 of the book.

Remark 5.12. Three claims about the world sit beside these results and none of them is used here: Landauer's principle (stated in its own chapter as an assumption), that a real device tells apart only finitely many states in a bounded space, and that a machine starts clean. Which reading counts as settling, and what the price list charges, are chosen.

6 The smallest one

Definition 6.1 (The latch)**.** [chosen] The *latch* is the machine with $S = \{\text{no}, \text{yes}\}$, one move stamp with $\delta(s, \text{stamp}) = \text{yes}$ for both s, the reading $r(s) = s$, the price $c(\text{stamp}) = 1$, and $S_0 = \{\text{no}\}$.

Argued in Chapter 4 of the book.

Proposition 6.2 (The smallest Thiele machine)**.** [shown]

(a) The latch is a Thiele machine, and two arrows meet at its move.

(b) Every Thiele machine has states s, t with $r(t) \not\leq r(s)$; so it has at least two states, and its reading is not constant.

(c) Let a Thiele machine have exactly two states. Let s be a start from which some run settles more (clause (ii)) and t the other state. Then $r(s) \leq r(t)$, $r(t) \not\leq r(s)$, and some move i has $\delta(s, i) = \delta(t, i) = t$. On the two-point reading t is the yes state, and the move i with its price is the latch.

(d) The toggle, two states and one move that swaps them, is not a Thiele machine for any reading, any price list, and any starts.

Proof. (a) Clause (i): $r(\delta(s, \text{stamp})) = \text{yes}$. Clause (ii): one stamp from no. Clause (iii): $c = 1$. Both states step to yes. (b) Clause (ii) gives s_0 and τ with $r(\text{run}(\tau, s_0)) \not\leq r(s_0)$; take $s = s_0$, $t = \text{run}(\tau, s_0)$; $s \neq t$ by reflexivity. (c) The run from s given by clause (ii) ends at a state whose reading is not at or below $r(s)$, so it ends at t, and $r(t) \not\leq r(s)$. Some step of it goes from s to t; call its move i. Clause (i) gives $r(s) \leq r(t)$ and $r(t) \leq r(\delta(t, i))$; if $\delta(t, i) = s$ then $r(t) \leq r(s)$, a contradiction; so $\delta(t, i) = t$. (d) For the states a, b of the toggle, clause (i) at a and at b gives $r(a) \leq r(b)$ and $r(b) \leq r(a)$. Then every state's reading is at or below every other's, no run settles more, and clause (ii) fails. □

Argued in Chapter 4 of the book.

Proposition 6.3 (Any machine can wear a coin)**.** [shown] *Let (S, I, δ) be any machine and $x \in S$ a state that some move lands on: $\delta(s_1, i_1) = x$ for some $s_1 \in S$, $i_1 \in I$. The* lift at x *has states $S \times \{\text{no}, \text{yes}\}$, the same moves, the step*

$$\delta'\big((s, b), i\big) = \big(\delta(s, i),\ b \vee [\delta(s, i) = x]\big),$$

where $\text{no} < \text{yes}$, $\vee$ is the larger of the two, and $[\delta(s, i) = x]$ is yes *exactly when $\delta(s, i) = x$; the reading $r'(s, b) = b$; the price $c'(i) = 1$ for every move; and the starts $S \times \{\text{no}\}$. Let $\pi(s, b) = s$.*

(a) The lift is a Thiele machine.

(b) Covering the coin gives back the machine: $\pi(\text{run}'(\tau, p)) = \text{run}(\tau, \pi(p))$ for every lifted state p and every list τ.

(c) The coin is not a function of what π shows: (x, no) is a start, (x, yes) is reached from the start (s_1, no), both have the picture x, and their readings differ. So no $g : S \to \{\text{no}, \text{yes}\}$ has $g(\pi(p)) = r'(p)$ for every lifted state p reached from a start.

In particular, every machine with at least one state and one move is what is left of a Thiele machine once the coin is covered and the price list dropped. For $S = \{*\}$ and one move $* \mapsto *$, the lift is the latch.

Proof. (a) $b \le b \vee [\cdot]$ gives clause (i). From (s_1, no) the move i_1 lands on (x, yes), which gives clause (ii). Every price is 1, so clause (iii) holds. (b) The first coordinate of δ' is δ; induction on τ. (c) Direct from (a) and the definitions. For the one-state machine, the states are $(*, \text{no})$ and $(*, \text{yes})$, and the one move sends both to $(*, \text{yes})$ at price 1. □

Argued in Chapter 4 of the book.

Argued in Chapter of the book.

Argued in Chapter 22 of the book.

Remark 6.4. A Turing machine read through whether it has stopped needs no extra coin: by Proposition 3.7 it is a Thiele machine once a price is put on its move.

Proposition 6.5 (Which question gets charged)**.** [shown]

(a) Let i be a move and s a state with $\delta(s, i) \neq s$. For the two-point reading $q(t) = \text{yes}$ if and only if $t = \delta(s, i)$, every price list that obeys the toll for q has $c(i) \ge 1$. So a price list that obeys the toll for every two-point reading on S charges at least one for every move that moves some state.

(b) In a Thiele machine, a move i with $c(i) = 0$ satisfies $r(\delta(s, i)) \le r(s)$ and $r(s) \le r(\delta(s, i))$ for every s. On the two-point reading, $r(\delta(s, i)) = r(s)$: a free move leaves the reading where it was.

Proof. (a) $q(s) = \text{no}$ and $q(\delta(s, i)) = \text{yes}$, so the toll applies. (b) The first inequality is the contrapositive of the toll, the second is clause (i). □

Argued in Chapter 4 of the book.

Proposition 6.6 (The cheapest price list)**.** [shown] *Let (S, I, δ) carry a reading r. Let $c^*(i) = 1$ if $r(\delta(s, i)) \not\le r(s)$ for some s, and $c^*(i) = 0$ otherwise. Then c^* obeys the toll, and $c^* \le c$ pointwise for every c that obeys the toll; so c^* is the unique least price list obeying the toll. For every c that obeys the toll, the price of every run is at least the number of its steps that settle more. On a two-point reading that never goes back, a run has at most one such step.*

Proof. The first two claims are the definition of the toll read move by move. Each step that settles more costs at least 1 and prices are in $\mathbb{N}$, so the sum is at least the count. On a two-point reading that never goes back, a step from no to yes is followed only by yes states. □

Argued in Chapter 4 of the book.

Remark 6.7. The toll is a floor on the price of a move, not a bill for what a step changes: under c^* the stamp costs one even when applied to yes.

7 More settled

Let $(R, \le)$ be a preorder. A *cup* on R is a map $u : R \to \{\text{no}, \text{yes}\}$ with $x \le y \Rightarrow u(x) \le u(y)$: once up, climbing never turns it down. The *cup at* $a \in R$ is $u_a(x) = \text{yes}$ if and only if $a \le x$. A *row* of k cups $u_1, \ldots, u_k$ sends x to its *arrangement* $(u_1(x), \ldots, u_k(x)) \in \{\text{no}, \text{yes}\}^k$, ordered coordinatewise. The row *tells places apart* when the arrangement is one-to-one, and *shows the order* when $x \le y$ holds exactly when the arrangement of x is at or below that of y. For a reading $r : S \to R$, the cup at a read on the machine is the two-point reading $u_a \circ r$.

Lemma 7.1. [shown] *Each u_a is a cup. If r satisfies clause (i) of Definition 2.2, each $u_a \circ r$ does too: each cup, read on the machine, is a two-point reading that never goes back.*

Proof. Transitivity of $\leq$. □

Argued in Chapter 5 of the book.

Proposition 7.2 (A row of cups on a straight line)**.** [shown]

(a) If $v_0 \leq v_1 \leq \cdots \leq v_n$ in $\{\text{no}, \text{yes}\}^k$ are pairwise distinct, then $n \leq k$.

(b) For each k, the arrangements c_j with the first j cups up, $j = 0, \ldots, k$, are $k+1$ pairwise distinct, increasing arrangements, and the row of k cups $u_j = u_{x_j}$, $j = 1, \ldots, k$, on the chain $x_0 < x_1 < \cdots < x_k$ sends x_j to c_j and shows the order.

(c) So a chain of n stages can be told apart by a row of k cups if and only if $k \geq n - 1$.

Proof. (a) Each $v_{t+1} \neq v_t$ with $v_t \leq v_{t+1}$ has at least one more coordinate up, and there are k coordinates. (b) $x_j \geq x_l$ exactly when $j \geq l$. (c) The stages form a chain, so their arrangements under any row of cups are increasing; if they are pairwise distinct, (a) gives $n - 1 \leq k$. (b) gives the converse. □

Argued in Chapter 5 of the book.

Proposition 7.3 (Cups fix the place exactly when the order is partial)**.** [shown] *For x, y in a preorder, $u_a(x) = u_a(y)$ for every a if and only if $x \leq y$ and $y \leq x$. So the cups at the points of R tell places apart if and only if $\leq$ is antisymmetric. Without antisymmetry: let $R = \{\ell, \rho\}$ with $\ell \leq \rho$, $\rho \leq \ell$, $\ell \neq \rho$, $S = R$, $r(s) = s$, and one move swapping ℓ and ρ. The reading never goes back, no cup ever changes, every price list obeys the toll, and clause (ii) fails, so this is not a Thiele machine.*

Proof. If all cups agree, $u_x(x) = \text{yes}$ gives $u_x(y) = \text{yes}$, that is $x \leq y$, and symmetrically. Conversely $x \leq y \leq x$ and transitivity make every cup agree. In the example every pair of places is related both ways, so no step settles more. □

Argued in Chapter 5 of the book.

Proposition 7.4 (When a step turns up a cup)**.** [shown] *For a reading $r : S \to R$, a state s and a move i, some cup $u_a \circ r$ goes from* no *at s to* yes *at $\delta(s, i)$ if and only if $r(\delta(s, i)) \not\leq r(s)$; the cup at $a = r(\delta(s, i))$ is one such cup. Clause (i) is not used.*

Proof. If $a \leq r(\delta(s, i))$ and $a \not\leq r(s)$, then $r(\delta(s, i)) \leq r(s)$ would give $a \leq r(s)$ by transitivity. Conversely the cup at $r(\delta(s, i))$ is up at $\delta(s, i)$ by reflexivity and down at s by hypothesis. □

Argued in Chapter 5 of the book.

Theorem 7.5 (The toll on the whole reading is the toll on each cup)**.** [shown] *For any machine (S, I, δ), reading $r : S \to R$ into a preorder, and price list c:*

(a) c obeys the toll for r if and only if, for every $a \in R$, c obeys the toll for the two-point reading $u_a \circ r$.

(b) If c obeys the toll, the price of every run is at least the number of its steps that turn up some cup; in particular a run at whose end some cup is up that was down at its start costs at least one.

Neither part uses clause (i), the shape of R, or the number of cups.

Proof. (a) By Proposition 7.4, a step turns up some cup exactly when it settles more for r, and a step turns up the cup at a exactly when it settles more for $u_a \circ r$. (b) Each such step settles more, so it costs at least one (Proposition 6.6). If the cup at a is down at the start of a run and up at its end, some step of the run turns it up. □

Argued in Chapter 5 of the book.

Proposition 7.6 (One cup is not enough)**.** [shown] *Let R be the chain $b < m < t$.*

(a) No single cup tells the three places apart.

(b) Let the record step by $g(b) = t$, $g(m) = m$, $g(t) = t$. No fixed $e \in R$ satisfies $g(x) = \sup\{x, e\}$ for all x: the rule "new place is the least place at or above the old place and a fixed event's place" does not carry this record.

Proof. (a) Three places, two values. (b) $\sup\{b, e\} = e$, so $g(b) = t$ forces $e = t$, and then $\sup\{m, t\} = t \neq m = g(m)$. □

Argued in Chapter 5 of the book.

Proposition 7.7 (How many cups a direction needs)**.** [shown] *Let $(R, \leq)$ be a finite partial order.*

(a) A row of k cups that tells places apart has $k \geq h$, where h is the number of steps $x_0 < x_1 < \cdots < x_h$ of a longest chain, and $2^k \geq |R|$. The cups at all points of R show the order.

(b) The fan: $R = \{\bot, p_1, p_2, p_3\}$ with $\bot < p_j$ and the p_j pairwise incomparable. Two cups tell places apart (the cups with up-sets $\{p_1, p_3\}$ and $\{p_2, p_3\}$), and no single cup does. A row that shows the order needs three cups, and the cups at p_1, p_2, p_3 do it.

(c) The crown: $R = \{a_1, a_2, b_1, b_2\}$ with $a_i < b_j$ for all i, j and no other strict relations. Telling places apart needs three cups, and the cups with up-sets $\{a_1, b_1, b_2\}$, $\{a_2, b_1, b_2\}$, $\{b_2\}$ do it. Showing the order needs four, and the cups at the four points do it.

Proof. (a) A chain's arrangements are increasing and pairwise distinct, so Proposition 7.2(a) gives $h \leq k$; the arrangement is one-to-one into a set of 2^k. For the cups at all points, $u_a(x) \leq u_a(y)$ for every a means that every $a \leq x$ has $a \leq y$, which by reflexivity and transitivity holds exactly when $x \leq y$. (b) The arrangements are $00, 10, 01, 11$, and each up-set is closed upward. Showing the order maps p_1, p_2, p_3 to three pairwise incomparable elements of $\{\text{no}, \text{yes}\}^k$; $\{\text{no}, \text{yes}\}^2$ has no three. (c) With two cups the four arrangements are all of $\{\text{no}, \text{yes}\}^2$. Nothing but 00 lies at or below 00, while each b_j lies above two other places, so 00 belongs to some a_i. Likewise 11 belongs to some b_j. The other a and the other b take the two middle arrangements, which are incomparable, while $a < b$ requires the other b to lie above the other a. The three given cups give the distinct arrangements $100, 010, 110, 111$, and each up-set is closed upward. To show the order with three cups, a_1, a_2 must go to incomparable arrangements, so their least upper bound

in $\{\text{no}, \text{yes}\}^3$ has at least two coordinates up, and b_1, b_2 must be incomparable arrangements above it. Above an arrangement with at least two coordinates up, the arrangements form a chain, a contradiction. □

Argued in Chapter 5 of the book.

Theorem 7.8 (The toll on any direction follows from the chairs)**.** [shown] *Let (S, I, δ) carry a reading $r : S \to R$ into a preorder, and let s, i satisfy $r(\delta(s,i)) \not\leq r(s)$. Let $a = r(\delta(s,i))$, and suppose the set $Y_a = \{t \in S : a \leq r(t)\}$ of states with the cup at a up is finite. Then two arrows meet at i, or some $t \in Y_a$ has $\delta(t,i) \notin Y_a$. If moreover r satisfies clause (i), two arrows meet at i. Hence, if clause (i) holds, every Y_a is finite, and c follows Bennett's rule, then c obeys the toll for r.*

Proof. The cup at a is a two-point reading with done set Y_a, down at s and up at $\delta(s,i)$ (Proposition 7.4). Proposition 5.3 applies to it. Under clause (i) the cup never goes back (Lemma 7.1), so no $t \in Y_a$ leaves Y_a, and two arrows meet. Bennett's rule then charges i. □

Argued in Chapter 5 of the book.

Remark 7.9. The three doors of Proposition 5.7 stay open on any direction: on $S = R = \mathbb{N}$ with its usual order, $r(n) = n$ and the free move $n \mapsto n+1$, nothing meets and every step turns up a new cup, with every Y_a infinite; the coin and the stamp at zero are already two-point readings. No extra condition is needed for places that are not comparable: every cup has its own set of chairs.

8 Looking isn't keeping

For a move i and a set $D \subseteq S$, write $\delta(D,i) = \{\delta(s,i) : s \in D\}$, and for a list of moves τ, $\text{run}(\tau, D) = \{\text{run}(\tau, s) : s \in D\}$. The *pile* of i over $y \in S$ is $\{s \in S : \delta(s,i) = y\}$. For a price list $c : I \to \mathbb{N}$ and $\tau = i_1 \cdots i_n$, $c(\tau) = \sum_k c(i_k)$. The two rules on price lists used below are Bennett's rule (Definition 5.1) and the halving rule (Definition 5.8).

Proposition 8.1 (The demon on four states)**.** [shown] *Let $S = \{0,1\}^2$, a state written (x,d) with x the hidden bit and d the display. Let* $\text{MEASURE}(x,d) = (x, d \oplus x)$ *and* $\text{WIPE}(x,d) = (x,0)$*, where $\oplus$ is addition mod 2, and let the window be $\pi(x,d) = d$.*

(a) MEASURE *is one-to-one,* MEASURE $\circ$ MEASURE *is the identity, and on a blank display it copies the hidden bit:* $\text{MEASURE}(x,0) = (x,x)$.

(b) Two arrows meet at WIPE*:* $\text{WIPE}(x,0) = \text{WIPE}(x,1)$ *for each x. Every price list obeying Bennett's rule has $c(\text{WIPE}) \geq 1$.*

(c) The price list $c(\text{MEASURE}) = 0$, $c(\text{WIPE}) = 1$ obeys the halving rule.

(d) Under that price list, one MEASURE *from either start in $\{(0,0),(1,0)\}$ costs 0. A watcher who sees π at every state of the run and knows only that the start lies in $\{(0,0),(1,0)\}$ sees display 0 afterwards from $(0,0)$ and display 1 from $(1,0)$, so after the run exactly one candidate start remains. The set of states the machine can be in after the run,* $\text{MEASURE}(\{(0,0),(1,0)\}) = \{(0,0),(1,1)\}$*, still has two elements.*

(e) On the two states $(0,0),(1,1)$ reached in (d), WIPE *is one-to-one; its images are $(0,0)$ and $(1,0)$.*

So a price list that obeys the halving rule can price at zero a step that narrows a watcher's candidates; the move every such price list must charge is the wipe. A blank display returned to blank by a second MEASURE *is the borrowed page handed back.*

Proof. (a) MEASURE sends $00 \mapsto 00$, $01 \mapsto 01$, $10 \mapsto 11$, $11 \mapsto 10$, a permutation, and $d \oplus x \oplus x = d$. (b) is read off the definition. (c) Every pile of MEASURE has one state and every pile of WIPE has at most two. (d) and (e) are read off the table. □

Argued in Chapter 6 of the book.

Proposition 8.2 (The machine's own row)**.** [shown] *If c obeys the halving rule, then for every finite $D \subseteq S$ and every list of moves τ,*

$$|D| \;\le\; 2^{c(\tau)}\,|\mathrm{run}(\tau, D)|.$$

Proof. By induction on τ. For the empty list both sides are $|D|$. For $\tau = i\tau'$, $|D| \le 2^{c(i)}|\delta(D,i)| \le 2^{c(i)}2^{c(\tau')}|\mathrm{run}(\tau', \delta(D,i))| = 2^{c(\tau)}|\mathrm{run}(\tau, D)|$. □

Argued in Chapter 6 of the book.

Remark 8.3. No bound of this form holds for a watcher's set of candidate starts: in Proposition 8.1(d) a run of cost 0 takes it from two elements to one, while the machine's set stays at two.

Proposition 8.4 (Learning after the first look)**.** [shown] *Let $f : S \to S$ and $\pi : S \to B$. Say* f separates after the first look *when there are $s \neq s'$ with $\pi(s) = \pi(s')$ and some $n \ge 1$ with $\pi(f^n(s)) \neq \pi(f^n(s'))$.*

(a) If $|S| \le 2$, no f and no π separate after the first look.

(b) On $S = \{U, V, W\}$ with $f(U) = V$, $f(V) = W$, $f(W) = U$ and $\pi(s) =$ yes exactly when $s = W$, f separates after the first look, with $s = U$, $s' = V$, $n = 1$. This f is one-to-one, so the price list $c = 0$ obeys the halving rule for it.

Proof. (a) If $s \neq s'$ and $|S| \le 2$, then $S = \{s, s'\}$, and $\pi(s) = \pi(s')$ makes π constant on S, so $\pi(f^n(s)) = \pi(f^n(s'))$ for every n. (b) $\pi(U) = \pi(V) =$ no, $\pi(f(U)) = \pi(V) =$ no, $\pi(f(V)) = \pi(W) =$ yes. □

Argued in Chapter 6 of the book.

Proposition 8.5 (Finishing together)**.** [shown] *Let $f : S \to S$, let $H \subseteq S$ be the finished states, let $A \subseteq S$ be a set of N starts and $T \in \mathbb{N}$. Suppose that for every $a \in A$, $f^t(a) \notin H$ for $t < T$ and $f^T(a) \in H$, and that for every $t < T$ and all $a, a' \in A$, $f^t(a) \neq f^t(a')$ implies $f^{t+1}(a) \neq f^{t+1}(a')$. Then the $N(T+1)$ states $f^t(a)$, $a \in A$, $0 \le t \le T$, are pairwise distinct. In particular $|S| \ge N(T+1)$.*

Proof. Same column: for $a \neq a'$, $f^0(a) \neq f^0(a')$, and the hypothesis carries this up to $t = T$. Different columns: if $f^t(a) = f^{t'}(a')$ with $t < t'$, apply $f^{T-t'}$ to both sides: $f^{t+T-t'}(a) = f^T(a') \in H$ with $t + T - t' < T$, which contradicts the first hypothesis for a. □

Argued in Chapter 6 of the book.

For a random variable X with finitely many values, $\mathrm{H}(X) = -\sum_x \Pr[X = x] \log_2 \Pr[X = x]$; $\mathrm{H}(X \mid Y) = \sum_y \Pr[Y = y]\,\mathrm{H}(X \mid Y = y)$; and $\mathrm{I}(X;Y) = \mathrm{H}(X) - \mathrm{H}(X \mid Y)$.

Proposition 8.6 (The certainty count)**.** [shown] *Let S be finite, let the start X be a random state of S with any distribution, let τ be a list of moves and $E = \mathrm{run}(\tau, X)$ the end state. Let $W = w(X)$ for a function w: what a watcher sees along the run, which depends only on the start.*

(a) $\mathrm{H}(X) = \mathrm{H}(X \mid E) + \mathrm{H}(E)$*: what was not known about the start is split into the part the run destroyed and the part the end state holds.*

(b) If c obeys the halving rule, $\mathrm{H}(X \mid E) \le c(\tau)$*.*

(c) $\mathrm{I}(W; X) \le \mathrm{I}(W; E) + \mathrm{H}(X \mid E)$*. Under the halving rule, a run with $c(\tau) = 0$ has* $\mathrm{I}(W; X) = \mathrm{I}(W; E)$*: everything the watcher learned about the start is still held in the end state.*

Proof. (a) E is a function of X, so $\mathrm{H}(X) = \mathrm{H}(X, E) = \mathrm{H}(E) + \mathrm{H}(X \mid E)$. (b) Given $E = e$, X lies in the pile of the run over e. By the halving rule each pile of a move i has at most $2^{c(i)}$ states, so the pile of the run has at most $\prod_k 2^{c(i_k)} = 2^{c(\tau)}$ states, and the entropy of a distribution on a set of size m is at most $\log_2 m$. (c) Since E is a function of X, $\mathrm{I}(W; X) = \mathrm{I}(W; X, E) = \mathrm{I}(W; E) + \mathrm{I}(W; X \mid E)$ and $\mathrm{I}(W; X \mid E) \le \mathrm{H}(X \mid E)$. With $c(\tau) = 0$, (b) gives $\mathrm{H}(X \mid E) = 0$, so $\mathrm{I}(W; X) \le \mathrm{I}(W; E)$; the reverse inequality holds because E is a function of X. □

Argued in Chapter 6 of the book.

Remark 8.7. None of this sets a quantity of heat. The halving rule is a lower bound on prices: a price list may also charge MEASURE, and which step a real device pays at is not settled here. The watcher's own notes need a blank page of their own. If the start is known for certain, $\mathrm{H}(X) = 0$ and Proposition 8.6 gives no floor.

9 Looking through a window

Fix a window $\pi : S \to B$. A *question* is a function $q : S \to A$. A *reader* for q through π is a function $g : B \to A$ with $g(\pi(s)) = q(s)$ for every $s \in S$.

Proposition 9.1 (A reader exists exactly when the question is constant on piles)**.** [shown] *A reader for q through π exists if and only if $q(s) = q(s')$ whenever $\pi(s) = \pi(s')$.*

Proof. If g is a reader and $\pi(s) = \pi(s')$, then $q(s) = g(\pi(s)) = g(\pi(s')) = q(s')$. Conversely, for b in the image of π let $g(b) = q(s)$ for any s with $\pi(s) = b$, which does not depend on the choice; let g be arbitrary off the image. □

Argued in Chapter 7 of the book.

Remark 9.2. The reader exists; nothing is claimed about computing it. Evaluating $g(b)$ may need a state behind b, and finding one may be hard or impossible to carry out.

Example 9.3 (Four readers)**.** [shown] Let a state be a pair (position, flag) with position in $\{0, 1\}$ and flag in $\{\mathrm{down}, \mathrm{up}\}$, and let π give the position. From $(0, \mathrm{down})$ one move goes to $(1, \mathrm{up})$ and another to $(1, \mathrm{down})$. There are exactly four functions $\{0, 1\} \to \{\mathrm{down}, \mathrm{up}\}$, and none gives the flag on both results, since both show position 1.

Argued in Chapter 7 of the book.

Proposition 9.4 (Same picture stays the same picture)**.** [shown] *Let $J \subseteq I$ be a set of moves such that for each $j \in J$ there is $h_j : B \to B$ with $\pi(\delta(s, j)) = h_j(\pi(s))$ for all s. If $\pi(s) = \pi(s')$, then for every list τ of moves in J the two runs from s and s' along τ show the same picture at every step. Consequently no function of the whole sequence of pictures (the album) answers a question that differs on s and s'.*

In particular, let $S = T \times K$ (tiles, stamp), $\pi(t, k) = t$, and let each tile move have the form $(t, k) \mapsto (h_j(t), k)$. Two boxes (t, k) and (t, k') with $k \neq k'$ keep equal tiles and keep stamps k and k' along every list of tile moves, and no reader of the album recovers the stamp.

Proof. By induction on τ: equal pictures before a move j give equal pictures $h_j(\pi(s))$ after it. The album of each run is then the same, and Proposition 9.1 applies with the album as the window. □

Argued in Chapter 7 of the book.

Remark 9.5. The hypothesis is that both boxes receive the same list of moves. If a move is chosen by reading the stamp, the stamp can reach the tiles, and the proposition does not apply.

Proposition 9.6 (No checker of a shared transcript)**.** [shown] *Let $\mathcal{T}$ be a set of transcripts, $\models \subseteq S \times \mathcal{T}$ a relation ($s \models t$: the state s can stand behind t), and P a claim on S. A checker $V : \mathcal{T} \to \{\text{no}, \text{yes}\}$ is* sound *when $V(t) = \text{yes}$ and $s \models t$ imply $P(s)$, and* complete *when $P(s)$ and $s \models t$ imply $V(t) = \text{yes}$. If some t has $s_1 \models t$ and $s_2 \models t$ with $P(s_1)$ and not $P(s_2)$, then no checker is both sound and complete, and every sound checker answers no on t.*

Proof. Completeness at s_1 gives $V(t) = \text{yes}$; soundness at s_2 then gives $P(s_2)$. □

Argued in Chapter 7 of the book.

Remark 9.7. In the chapter's instance, a run that presses the stamp and counts 1 and a run that only moves tiles and counts 0 go from the same start to the same picture; the transcript is the pair of pictures, and the claim is "the count is 1". Adding parts to the transcript that are equal on both runs leaves the hypothesis in place.

Lemma 9.8 (The running bit)**.** [shown] *Let $p_1, p_2, \cdots \in \{\text{no}, \text{yes}\}$ be the outcomes of successive checks, $b_0 = \text{yes}$ and $b_k = \text{yes}$ exactly when $b_{k-1} = \text{yes}$ and $p_k = \text{yes}$. Then $b_k = \text{yes}$ if and only if $p_1 = \cdots = p_k = \text{yes}$, and once $b_k = \text{no}$, $b_{k'} = \text{no}$ for all $k' \geq k$.*

Proof. Induction on k. □

Argued in Chapter 7 of the book.

Fix a reading $r : S \to (R, \leq)$. A step (s, i) *settles* when $r(\delta(s, i)) \not\leq r(s)$.

Definition 9.9 (Clerk prices)**.** [chosen] A *clerk price* is a function $p : B \times B \to \mathbb{N}$, charged on a step as $c_p(s, i) = p(\pi(s), \pi(\delta(s, i)))$. It *meets the floor* when $c_p(s, i) \geq 1$ on every settling step, *never overcharges* when $c_p(s, i) = 0$ on every step that does not settle, and is *exact* when it does both. The *tray* of (b, b') is the set of steps (s, i) with $\pi(s) = b$ and $\pi(\delta(s, i)) = b'$. A *collision* is a tray that holds a settling step and a step that does not settle. A step that does not settle is *shadowed* when its tray holds a settling step.

Theorem 9.10 (A window with a clash cannot price exactly)**.** [shown]

(a) If there is a collision, every clerk price that meets the floor charges at least 1 on some step that does not settle; so no clerk price is exact.

(b) *If the window shows the reading, $r = \rho \circ \pi$ for some $\rho : B \to R$, then $p(b, b') = 1$ if $\rho(b') \not\leq \rho(b)$ and 0 otherwise is exact, and there is no collision.*

(c) *An exact clerk price exists if and only if there is no collision.*

(d) *(Least overcharge.) For every clerk price p that meets the floor and every run, the total that p charges along the run is at least the number of shadowed steps in the run.*

(e) *(A price that reaches it.) The clerk price $p_\exists(b, b') = 1$ when the tray of (b, b') holds a settling step, and 0 otherwise, meets the floor and charges a step that does not settle 1 if it is shadowed and 0 if not. Along every run its overcharge equals the number of shadowed steps.*

(f) *The clerk price $p_\forall(b, b') = 1$ when the tray of (b, b') is nonempty and all its steps settle, and 0 otherwise, never overcharges. $p_\exists$ and $p_\forall$ agree on every nonempty tray if and only if there is no collision.*

Proof. (a) Let (s_1, i_1) settle and (s_2, i_2) not settle in one tray (b, b'). Then $c_p(s_2, i_2) = p(b, b') = c_p(s_1, i_1) \geq 1$. (b) $c_p(s, i) = 1$ exactly when $r(\delta(s, i)) \not\leq r(s)$; a collision would put a settling and a non-settling step on the same pair $(\rho(b), \rho(b'))$. (c) One direction is (a). Without a collision every nonempty tray is all settling or all not settling, and $p_\exists$ is exact. (d) Each shadowed step lies in a tray with a settling step, so p charges it at least 1; sum along the run. (e) and (f) follow from the definitions of the trays. □

Argued in Chapter 7 of the book.

Remark 9.11. Seeing the whole state and the move prices exactly. A window with no collision prices exactly without showing the reading; a later chapter has one that shows when the record moves and not where it is. Three ways past a window are named in the chapter (the whole state, a promise mark, a question answered); each rests on trusting something, and none is shown here to resist a cheating party.

10 Established

The condition of Definition 2.4 is used throughout.

Proposition 10.1 (Two computers, one answer)**.** [shown] *Let $S \neq \emptyset$, let* $\mathrm{flag} : S \to \{\mathrm{no}, \mathrm{yes}\}$, *and let* $\mathrm{set} : S \times \{\mathrm{no}, \mathrm{yes}\} \to S$ *satisfy* $\mathrm{flag}(\mathrm{set}(s, b)) = b$ *and* $\pi(\mathrm{set}(s, b)) = \pi(s)$ *for all s, b: the flag can be set on its own, and the window drops it. Then there is no reader $g : B \to \{\mathrm{no}, \mathrm{yes}\}$ with $g(\pi(s)) = \mathrm{flag}(s)$ for all s. In particular two end states* $\mathrm{set}(s_0, \mathrm{yes})$ *and* $\mathrm{set}(s_0, \mathrm{no})$ *show the same picture and differ only in the flag.*

Proof. For any s_0, $g(\pi(s_0)) = g(\pi(\mathrm{set}(s_0, \mathrm{yes}))) = \mathrm{yes}$ and likewise $g(\pi(s_0)) = \mathrm{no}$. □

Argued in Chapter 8 of the book.

Proposition 10.2 (Room for the sheet)**.** [shown] *Let V be a finite set of values, C a set of checks, and for $v \in V$ let $P(v) \subseteq C$ be the finite set of checks that pass on v, $m(v) = |P(v)|$. Suppose a machine has readings* $\mathrm{val} : S \to V$ *and* $\mathrm{tick}_\kappa : S \to \{\mathrm{no}, \mathrm{yes}\}$ *for $\kappa \in C$, and for every v and every $X \subseteq P(v)$ a state s with* $\mathrm{val}(s) = v$ *and* $\{\kappa : \mathrm{tick}_\kappa(s) = \mathrm{yes}\} = X$. *Then*

$$|S| \;\geq\; \sum_{v \in V} 2^{m(v)}.$$

The machine whose states are the pairs (v, X) with $X \subseteq P(v)$, with a move writing w that goes to $(w, \emptyset)$ and a move checking κ that goes from (v, X) to $(v, X \cup \{\kappa\})$ when $\kappa \in P(v)$ and stays put otherwise, has exactly that many states.

Proof. Two pairs $(v, X) \neq (v', X')$ differ in val or in some tick_κ, so the states that realize them differ. □

Argued in Chapter 8 of the book.

Example 10.3 (The check is a merge)**.** [shown] Take one claim and three states: F (false), TU (true, unchecked), TC (true, checked). The check sends $F \mapsto F$, $TU \mapsto TC$, $TC \mapsto TC$. Two arrows meet at the check, above the same value "true", so every price list obeying Bennett's rule charges the check at least 1.

Argued in Chapter 8 of the book.

Proposition 10.4 (A file of answers needs endless room)**.** [shown] *Suppose that after n rounds, for every history $h \in \{0,1\}^n$ (round k checked or not), a machine has a run ending in a state s_h, all with the same list value, and readings $r_1, \ldots, r_n : S \to \{\text{no}, \text{yes}\}$ with $r_k(s_h) = h_k$. Then the 2^n states s_h are pairwise distinct. If this holds for every n, S is infinite. A machine that files each round's answer and resets its page and marker meets this for every n, with 2^n distinct end states above one final list.*

Proof. Histories $h \neq h'$ differ at some k, so $r_k(s_h) \neq r_k(s_{h'})$. □

Argued in Chapter 8 of the book.

Remark 10.5. The question "was anything ever checked" is the disjunction of the file. As a reading it is a single box that only goes up, which is a reading of a Thiele machine.

Example 10.6 (The sorter and the copier)**.** [shown] Let L be a set of lists and $\text{ok} : L \to \{\text{no}, \text{yes}\}$ the claim "in order". States are triples (v, m, b) with $v \in L$, a mark $m \in \{0, 1\}$ and a box $b \in \{\text{no}, \text{yes}\}$; the reading is b; the starts are the states $(v, 0, \text{no})$. Moves and prices: WRITE_w sends (v, m, b) to $(w, 0, b)$ at price 0 (ordinary work; it touches the list); CHECK sends (v, m, b) to $(v, 1, b)$ if $\text{ok}(v)$ and leaves it otherwise, at price 1, and passes at (v, m, b) exactly when $\text{ok}(v)$; RAISE sends $(v, 1, b)$ to $(v, 1, \text{yes})$ and leaves $(v, 0, b)$, at price 1. Suppose some list is in order.

(a) This is a Thiele machine, and the box is established for "in order".

(b) CHECK makes two arrows meet: $(v, 0, b)$ and $(v, 1, b)$ both go to $(v, 1, b)$ when $\text{ok}(v)$.

(c) Let $u \in L$ and $w \in L$ with $\text{ok}(w)$. The sorter's run $\text{WRITE}_w, \text{CHECK}, \text{RAISE}$ from $(u, 0, \text{no})$ ends at $(w, 1, \text{yes})$ at total price 2. The copier's run $\text{WRITE}_w, \text{RAISE}$ from $(u, 0, \text{no})$ ends at $(w, 0, \text{no})$ at total price 1. The lists agree; the box and the count differ. A window that shows only the list gives the same picture for both ends.

Proof. (a) The box never goes down; the sorter's run raises it; the only move that raises it is RAISE, priced 1. From a start the mark is 0; it becomes 1 only by a passing CHECK and returns to 0 at every WRITE, so every raising RAISE is preceded by a passing check with no write between. (b) and (c) are read off the moves. □

Argued in Chapter 8 of the book.

Proposition 10.7 (The toll alone does not establish)**.** [shown] *There is a Thiele machine whose box is not established for any claim: states (v, b) with $v \in L$, reading b, starts (v, no), and one move* RAISE *that sends (v, b) to (v, yes) at price 1, besides price-0 writes of the list. From a clean start the run* RAISE *turns the box up with no check before it, and the toll is paid.*

Proof. Clauses (i) to (iii) of Definition 2.2 hold by inspection. The raise in the one-step run is preceded by no step at all. □

Argued in Chapter 8 of the book.

Proposition 10.8 (What a raised established box tells)**.** [shown] *Let the box be established for κ. On every run from a clean start that reaches a state with the box up, some step earlier in the run is a passing check of κ, followed by no move that touches what κ is about until a step that turns the box up. If moreover a passing check of κ at a state implies a predicate P there, and the check moves and every move that does not touch what κ is about keep P true, then P holds at the state from which that step turns the box up. Nothing follows about P at later states.*

Proof. The box is down at the clean start and up later, so some step turns it up; the definition gives the check before it. P holds after the check and is kept by every step up to the raise. In Example 10.6, a WRITE of a list not in order after the raise leaves the box up and the claim false. □

Argued in Chapter 8 of the book.

Remark 10.9. That a check means what it says and could have failed is a fact about the checker, not a condition on the machine (Definition 2.4). The guarantee covers runs from a clean start along the machine's own step; a start that is not clean, or a change made outside the step, is outside it. A record and a count that no step reads, other than to update them, can be erased without changing what the machine computes.

11 Settled

Definition 11.1 (A settled question)**.** [chosen] A *question* on the states is a function $Q : S \to \{\text{no}, \text{yes}\}$. It *sorts* S when it is not constant: then $\{Q^{-1}(\text{no}), Q^{-1}(\text{yes})\}$ is a partition of S into two nonempty blocks. It has a *one-way door* when $Q(s) = \text{yes}$ implies $Q(\delta(s, i)) = \text{yes}$ for every $s \in S$ and $i \in I$. A *settled question* is a sorting question with a one-way door.

Proposition 11.2 (Settling is a sorting with a one-way door; closed is not true)**.** [shown; the reading is mine]

(a) A question Q has a one-way door if and only if Q, as a two-point reading, meets clause (i) of Definition 2.2; the yes side is closed under every move.

(b) The door places no condition linking the yes side to any claim. In Example 10.6, replace CHECK *by a check that passes at every state and sets the mark. The result is a Thiele machine whose box is established, in the sense of Definition 2.4, for the claim "in order", and the run* $\text{WRITE}_u, \text{CHECK}, \text{RAISE}$ *with* $\text{ok}(u) = \text{no}$ *ends with the box up and the claim false.*

Proof. (a) On $\{\text{no}, \text{yes}\}$ with $\text{no} < \text{yes}$, $Q(s) \le Q(\delta(s, i))$ fails exactly when $Q(s) = \text{yes}$ and $Q(\delta(s, i)) = \text{no}$. (b) The argument of Example 10.6(a) uses nothing about when the check passes. □

Argued in Chapter 9 of the book.

Remark 11.3. Proposition 10.8 says how far a closed question reaches toward truth: through a check that passes only when the claim holds, and no further. After a raise the record stays up about the version checked; a later change of the thing, or a second record of withdrawal, does not bring it down.

Example 11.4 (Three ballots)**.** [shown] A payment is owed exactly when a contract was signed and a deadline was missed. Three voters answer (signed, missed) as (yes, yes), (yes, no), (no, yes), and each applies the rule. A majority says signed, a majority says missed, and a majority says not owed. The general impossibility for combining judgments on connected questions is List and Pettit's [cited].

Argued in Chapter 9 of the book.

Proposition 11.5 (A chain of reasons)**.** [shown] *Let n be a partial function from a set X to itself (each reason has at most one next reason), and $x_0 \in X$, $x_{k+1} = n(x_k)$ while defined. Exactly one of the following holds: the sequence stops; some term repeats an earlier one; the sequence is defined forever with all terms distinct. If X is finite, the third does not occur.*

Proof. If the sequence does not stop and no term repeats, it is defined forever with distinct terms. A finite X has no infinite sequence of distinct elements. □

Argued in Chapter 9 of the book.

12 What structure is

Proposition 12.1 (The diagonal)**.** [shown] *There are no positive integers p, q with $p^2 = 2q^2$: the diagonal of a square is not a ratio of whole numbers to its side.*

Proof. Suppose there are, and take p/q with no common factor. p^2 is even, and the square of an odd number is odd, so $p = 2r$. Then $4r^2 = 2q^2$, so $q^2 = 2r^2$ is even and q is even, against the choice of p, q. □

Argued in Chapter 10 of the book.

Definition 12.2 (Structure, relative to a frame)**.** [chosen (an overlay of the sources)] A *pile* is a set X. A *frame* on X is a set G of one-to-one maps of X onto itself (the chosen changes) that contains the identity and is closed under composition and inverses. Relative to G, the relations that count are the relations $R \subseteq X^n$ with $g(R) = R$ for every $g \in G$, and $x, y \in X$ are *the same* when $g(x) = y$ for some $g \in G$. Two piles with named relations carry the same structure when a one-to-one map of one onto the other carries each named relation onto the relation of the same name. There is no structure of a pile without a frame.

Argued in Chapter 10 of the book.

Proposition 12.3 (Sameness under a frame sorts the pile)**.** [shown] *For a frame G on X, "$g(x) = y$ for some $g \in G$" is an equivalence relation on X.*

Proof. The identity gives reflexivity, inverses give symmetry, composition gives transitivity. □

Argued in Chapter 10 of the book.

Proposition 12.4 (Sameness and sorting are one object)**.** [shown (a textbook fact)] *On a set X, sending an equivalence relation $\sim$ to its classes $\{\{y : y \sim x\} : x \in X\}$ is a one-to-one correspondence between equivalence relations on X and partitions of X into nonempty blocks; the inverse sends a partition to the relation "in the same block".*

Proof. The classes are nonempty ($x \sim x$) and cover X. If z lies in the classes of x and of y, then $x \sim z \sim y$, so $x \sim y$ and the two classes are equal: blocks never half overlap. "Same block" is reflexive, symmetric and transitive because each thing lies in exactly one block. Going one way and back returns the relation, and returns the partition. □

Argued in Chapter 10 of the book.

Proposition 12.5 (An order is a sorting with one-way doors)**.** [shown (a textbook fact)] *Let $\leq$ be a preorder on R (reflexive and transitive), and let $x \equiv y$ when $x \leq y$ and $y \leq x$.*

(a) $\equiv$ is an equivalence relation (the buckets).

(b) $\leq$ respects buckets: $x \equiv x'$, $y \equiv y'$ and $x \leq y$ imply $x' \leq y'$.

(c) Between different buckets $\leq$ goes one way: $x \leq y$ and $x \not\equiv y$ imply $y \not\leq x$. So the order induced on buckets is a partial order.

(d) Conversely, a partition of R with a partial order on its blocks gives, by $x \leq y$ when $\text{block}(x) \leq \text{block}(y)$, a preorder whose buckets are the blocks.

Removing the doors leaves the partition.

Proof. (a) and (b) are reflexivity and transitivity. (c) If also $y \leq x$, then $x \equiv y$. (d) Reflexivity and transitivity pass from the order on blocks; $x \leq y$ and $y \leq x$ give equal blocks by antisymmetry. □

Argued in Chapter 10 of the book.

Definition 12.6 (Vocabulary and the ruler)**.** [chosen] A *vocabulary* is a finite list of names, each with an arity $n \geq 0$ (how many things it looks at at once) and a nonempty set of answers. A *structure* for it on a pile X gives each name of arity n with answers A a function $X^n \to A$. A name *tells x from y* when replacing x by y in some one position of some argument tuple changes its answer; a structure *sorts X* when some name tells two things of X apart. The *ruler* compares vocabularies first by the number of names, then by the largest arity, then by the largest number of answers.

Argued in Chapter 10 of the book.

Proposition 12.7 (The smallest structure that sorts)**.** [shown] *Let $|X| \geq 2$. Under the ruler, the least vocabulary that has a structure sorting X is one name of arity one with two answers, that is, one yes-or-no question, interpreted as any question that is not constant on X.*

Proof. With no names nothing is told apart. A name of arity 0 has no position to replace. A name of arity one with one answer is constant. A name of arity one with two answers tells x from y exactly when its answers on them differ, and any nonconstant question on X does so. □

Argued in Chapter 10 of the book.

Proposition 12.8 (A question is a split with a label)**.** [shown] *Sending a nonconstant* $Q : X \to \{\text{no}, \text{yes}\}$ *to* $\{Q^{-1}(\text{no}), Q^{-1}(\text{yes})\}$ *maps the nonconstant questions onto the partitions of* X *into two nonempty blocks, two to one: the two questions over a partition differ by which block is called yes.*

Proof. A partition $\{Y, Z\}$ comes from exactly the two questions with yes side Y and with yes side Z. □

Argued in Chapter 10 of the book.

Proposition 12.9 (The walking arrow)**.** [chosen ruler; shown under it] *Let* $\to$ *be any relation on* S *(the machine's moves) and* side $: S \to \{\text{NO}, \text{YES}\}$*. Say* side *lays the moves onto a picture* $P \subseteq \{\text{NO}, \text{YES}\}^2$ *when* $s \to s'$ *implies* $(\text{side}(s), \text{side}(s')) \in P$*.*

(a) The moves lay onto the walking arrow $\{(\text{NO}, \text{NO}), (\text{NO}, \text{YES}), (\text{YES}, \text{YES})\}$ *if and only if no move goes from a YES state to a NO state: a split with a one-way door.*

(b) The moves lay onto two dots with loops only, $\{(\text{NO}, \text{NO}), (\text{YES}, \text{YES})\}$*, if and only if no move crosses the split.*

(c) Measured by fewest dots, then fewest arrows between different dots, the smallest picture with a direction (an arrow between different dots) is the walking arrow: one dot has no such arrow, and two dots need at least one.

Proof. (a) The only pair missing from the walking arrow is (YES, NO). (b) The missing pairs are the two crossings. (c) is counting. □

Argued in Chapter 10 of the book.

Example 12.10 (Three descriptions of a row)**.** [shown] Rows have 32 binary digits, a length both sides know. A description starts with one digit: 0 for the row written out ($1 + 32 = 33$ bits); 1 for a pattern, given as five digits for the block length minus one, the block itself, five digits for the number $e \leq 31$ of exceptions, and five digits for each exception's place in $\{0, \ldots, 31\}$, listed increasing; the block is repeated and cut at 32 digits, and the digit at each listed place is flipped. A pattern with block length ℓ and e exceptions costs $1 + 5 + \ell + 5 + 5e$ bits. For the row $(01)^{16}$:

(a) "repeat 01" costs 13; with e flipped digits it costs $13 + 5e$: 23 for $e = 2$, 33 (a tie with writing out) for $e = 4$, 38 for $e = 5$;

(b) flipping places $5, 11, 17, 23, 29$ gives the row $(010100)^5 01$, described by "repeat 010100" with no exceptions at $1 + 5 + 6 + 5 = 17$;

(c) the whole row as its own block costs $1 + 5 + 32 + 5 = 43$;

(d) with seven digits per exception place, three flips cost $8 + 5 + 21 = 34 > 33$, where the five-digit code gives $28 < 33$.

These numbers are lengths in this code only; they are not the price of a step.

Argued in Chapter 10 of the book.

Proposition 12.11 (Few short descriptions)**.** [shown] *In any code where each description prints one row, fewer than* 2^{n-k} *of the* 2^n *rows of length* n *have a description shorter than* $n - k$ *bits, a fraction below* 2^{-k}*.*

Proof. There are $\sum_{j<n-k} 2^j = 2^{n-k} - 1$ binary strings shorter than $n-k$, and each prints at most one row. □

Argued in Chapter 10 of the book.

Proposition 12.12 (Buckets with and without the blocks)**.** [shown] *Let $\ell : X \to L$ assign each block a bucket. The step $x \mapsto (x, \ell(x))$, writing the label beside a kept block on a blank place, is one-to-one. The step $x \mapsto \ell(x)$, keeping only the label, makes two arrows meet if and only if two blocks share a bucket.*

Proof. The first coordinate recovers x. The second map is one-to-one exactly when ℓ is. □

Argued in Chapter 10 of the book.

Remark 12.13. A box that already reads yes can stay there forever under the identity move, which is one-to-one and never settles more: no merge and no fresh state. The merge-or-fresh count of the first three chapters concerns a box that starts at no and is turned to yes and kept. A grouping by itself supplies neither that run nor a price.

Remark 12.14 (Not claimed)*.* [not claimed] It is not claimed that nobody prices vouching for a grouping. The chapter reports only that its author found no source that does.

13 The counter

Definition 13.1 (A counter kept by a price list)**.** [chosen] A *counter machine* is a machine (S, I, δ) with a start s_0, a price list $c : I \to \mathbb{N}$, and a counter $\mu : S \to \mathbb{N}$ in the state with $\mu(s_0) = 0$ and $\mu(\delta(s,i)) = \mu(s) + c(i)$ for all s and i. A state is *reachable* when it is $\mathrm{run}(\tau, s_0)$ for some list τ.

Proposition 13.2 (The total is the charges)**.** [shown] *In a counter machine, $\mu(\mathrm{run}(\tau, s)) = \mu(s) + c(\tau)$ for every state s and list τ; so μ never decreases along a run. For any $m : S \to \mathbb{N}$, $m(\mathrm{run}(\tau, s)) = m(s) + c(\tau)$ for all s, τ holds if and only if $m(\delta(s,i)) = m(s) + c(i)$ for all s, i.*

Proof. Induction on τ; the one-step rule is the case of a list of length one. □

Argued in Chapter 11 of the book.

Proposition 13.3 (Only one honest count)**.** [shown] *In a counter machine, let $B : S \to \mathbb{N}$ satisfy $B(s_0) = 0$ and $B(\delta(s,i)) = B(s) + c(i)$ for every reachable s and every i. Then $B(s) = \mu(s)$ for every reachable s. Both hypotheses are needed: $B = \mu + 1$ follows the price list and agrees nowhere; and a state no run reaches can carry any value.*

Proof. Induction on the run reaching s: both start at 0 and rise by $c(i)$ on each step. □

Argued in Chapter 11 of the book.

Proposition 13.4 (A counter in the state takes fresh states)**.** [shown] *In a counter machine:*

(a) if a run visits the same state twice, the moves between the two visits cost 0 *in total;*

(b) a step whose move costs at least 1 *lands on a state the run has not visited before;*

(c) *if S is finite, $c(i) = 0$ for every move i. So a counter machine with a move of price at least 1 has infinitely many states.*

Proof. (a) By Proposition 13.2 the counter rises by the cost of the moves between, and returns to its value. (b) The counter after the step exceeds its value at every earlier state of the run. (c) Take s with $\mu(s)$ largest; then $\mu(s) + c(i) = \mu(\delta(s,i)) \le \mu(s)$. □

Argued in Chapter 11 of the book.

Proposition 13.5 (Walking one machine into another)**.** [cited] *This is the form of Goguen's realization result used in the chapter. Let M_1 and M_2 have the same moves, starts s_1, s_2, and runs $\mathrm{run}_1, \mathrm{run}_2$. A map φ from the reachable states of M_1 to the states of M_2 with $\varphi(\mathrm{run}_1(w, s_1)) = \mathrm{run}_2(w, s_2)$ for every list w exists if and only if $\mathrm{run}_1(w, s_1) = \mathrm{run}_1(w', s_1)$ implies $\mathrm{run}_2(w, s_2) = \mathrm{run}_2(w', s_2)$. Such a φ then commutes with every move.*

Proof. Set $\varphi(\mathrm{run}_1(w, s_1)) = \mathrm{run}_2(w, s_2)$; the condition is exactly that this does not depend on w. Then $\varphi(\delta_1(\mathrm{run}_1(w, s_1), i)) = \mathrm{run}_2(wi, s_2) = \delta_2(\varphi(\mathrm{run}_1(w, s_1)), i)$. □

Argued in Chapter 11 of the book.

Example 13.6 (One switch against the whole list)**.** [shown] Let M_1 have states $\{\mathrm{down}, \mathrm{up}\}$ from down, with set $\mapsto$ up at price 1 and reset $\mapsto$ down at price 0; let M_2 keep the list of moves made. The lists set and reset set both bring M_1 to up and bring M_2 to different states, so no map as in Proposition 13.5 exists; the map from lists into M_2 exists and is unique. The toll is not used.

Argued in Chapter 11 of the book.

Proposition 13.7 (The potential method)**.** [cited] *This is the potential method of Tarjan, crediting Sleator. Let $c : S \times I \to \mathbb{N}$, $\Phi : S \to \mathbb{Z}$, and write $c(\tau, s)$ for the total price of the run from s along τ. If $c(s,i) + \Phi(\delta(s,i)) - \Phi(s) \ge 0$ for all s, i, then $c(\tau, s) \ge \Phi(s) - \Phi(\mathrm{run}(\tau, s))$ for every s and τ; conversely the bound for every run implies the one-step condition. If $c(s,i) + \Phi(\delta(s,i)) - \Phi(s) \le a$ for all s, i, then $c(\tau, s) \le a|\tau| + \Phi(s) - \Phi(\mathrm{run}(\tau, s))$.*

Proof. Sum the one-step inequality along the run; the potentials telescope. The converse is the run of length one. □

Argued in Chapter 11 of the book.

Example 13.8 (The stack of coins)**.** [shown] States are $s \in \mathbb{N}$ (coins on the stack). The move op_k takes $\min(k, s)$ coins off and puts one on, at price $\min(k, s) + 1$. With $\Phi(s) = s$, every step has $c + \Phi(\text{after}) - \Phi(\text{before}) = 2$, so every run of n moves from the empty stack costs $2n - \Phi(\text{end}) \le 2n$.

Argued in Chapter 11 of the book.

Theorem 13.9 (The toll as a potential; floor and ceiling)**.** [shown] *Let r be a two-point reading, $\Phi(s) = 1$ when $r(s) =$ no and $\Phi(s) = 0$ when $r(s) =$ yes, and let $c : S \times I \to \mathbb{N}$ depend on the state a step starts from. A step (s, i) is a* raise *when $r(s) =$ no and $r(\delta(s,i)) =$ yes.*

(a) *(Floor.) $c(s,i) + \Phi(\delta(s,i)) \ge \Phi(s)$ for all s, i if and only if $c(s,i) \ge 1$ on every raise. Then along every run the number of raises is at most the total price, and every run from a no state to a yes state costs at least 1.*

(b) *(Floor and ceiling.) The floor together with* $\Phi(s) \geq c(s,i) + \Phi(\delta(s,i))$ *for all* s, i *holds if and only if* $c(s,i) = 1$ *on every raise,* $c(s,i) = 0$ *on every other step, and no step goes from yes to no.*

(c) *(Per move.) If some move raises at some state, no price list* $c : I \to \mathbb{N}$*, the same at every state, meets both floor and ceiling.*

Proof. (a) $\Phi(s) - \Phi(\delta(s,i))$ is 1 on a raise, -1 on a step from yes to no, and 0 otherwise; prices are at least 0. Each raise then costs at least 1; sum along the run. (b) Floor and ceiling together say $c(s,i) = \Phi(s) - \Phi(\delta(s,i))$. A step from yes to no would need price -1; a raise gets 1; every other step gets 0. The converse is a check of the same three cases. (c) Let i raise at s, and $s' = \delta(s,i)$, a yes state. The ceiling at (s',i) gives $c(i) \leq 0 - \Phi(\delta(s',i)) \leq 0$, and the floor at (s,i) gives $c(i) \geq 1$. □

Argued in Chapter 11 of the book.

Corollary 13.10 (A window that leaves out the count)**.** [shown] *If two states have different counters and the same picture through* π*, no reader recovers the counter through* π*.*

Proof. Proposition 9.1 with $q = \mu$. □

Argued in Chapter 11 of the book.

Remark 13.11. Given its start and its price list, the counter is arithmetic. Nothing here says what the prices should be, or what a unit of spending was worth.

14 What a ticked box lets you lean on

Definition 14.1 (Entitlement)**.** [chosen] A run is *entitled to lean on* a claim κ when three things are on hand:

(i) an *object*: a component of the states of the run;

(ii) a *checked relation*: a step of the run that is a check relating that object to κ, with its result kept in the state;

(iii) a *paid event*: a step of the run that put the object in place and whose price is at least 1.

Entitlement is a statement about the run, made by whoever examines it afterwards. Nothing in the machine requires a later step to read the object or the box.

Argued in Chapter 12 of the book.

Remark 14.2. [not claimed] It is not claimed that a ticked box proves its claim true. An established box (Definition 2.4) says that a check of the claim passed before the raise, with nothing the claim is about changed in between. That the check says no whenever the claim is false is a fact about the checker, not a condition on the machine, and a check that says yes to everything meets the definition.

Argued in Chapter 12 of the book.

Narrowing and the strength of a test

Let L be a set of candidates and $o : L \to O$ an observation. The *test* of L is the set $o(L) = \{o(x) : x \in L\}$ of observations some candidate in L would show; the test passes v when $v \in o(L)$.

Lemma 14.3 (Narrower means a stronger test)**.** [shown] *Let $L' \subseteq L$. Then $o(L') \subseteq o(L)$, and $o(L') \subsetneq o(L)$ if and only if some $x \in L \setminus L'$ has $o(x) \neq o(y)$ for every $y \in L'$.*

Proof. The inclusion is immediate. The test of L passes an observation that the test of L' does not exactly when some $x \in L$ shows a value that no $y \in L'$ shows; such an x is not in L'. □

Example 14.4. Let L be the six orders of the cards $1, 2, 3$ and $L' = \{123, 132, 312\}$, the orders with 1 left of 2. For the observation "is 1 left of 2", $o(L') = \{\text{yes}\} \subsetneq \{\text{yes}, \text{no}\} = o(L)$. For the observation "is the first card a 3", $o(L') = o(L) = \{\text{yes}, \text{no}\}$, although three orders were dropped.

Argued in Chapter 12 of the book.

Lemma 14.5 (Halvings and question trees)**.** [shown] *Let C be a finite set of $N \geq 1$ candidates and let a binary tree of yes-or-no questions of depth d send each candidate to a leaf, with at most $M \geq 1$ candidates at each leaf. Then $N \leq 2^d M$, and*

$$d \;\geq\; \lceil \log_2 N \rceil - \lceil \log_2 M \rceil .$$

If a run has at least d steps each priced at least 1, its price is at least $\lceil \log_2 N \rceil - \lceil \log_2 M \rceil$.

Proof. A tree of depth d has at most 2^d leaves, so $N \leq 2^d M \leq 2^{d + \lceil \log_2 M \rceil}$, which gives $\lceil \log_2 N \rceil \leq d + \lceil \log_2 M \rceil$. The last sentence follows because the price of a run is the sum of the prices of its steps. □

Example 14.6. Sorting three cards by comparisons separates $3! = 6$ orders down to one, so $M = 1$ and $d \geq \lceil \log_2 6 \rceil = 3$. For twelve cards the bound is $\lceil \log_2 12! \rceil = 29$, and Wells's computer search found that 30 comparisons are needed [cited]: the bound is a floor and need not be reached.

Lemma 14.7 (Bounded share per test)**.** [shown] *Let $k \geq 0$ and let a run make t tests that narrow a list of candidates through sizes $L_0 \geq L_1 \geq \cdots \geq L_t \geq 1$, where each test keeps at least a share 2^{-k} of the list: $L_{j-1} \leq 2^k L_j$ for $1 \leq j \leq t$. Then*

$$\lceil \log_2 L_0 \rceil - \lceil \log_2 L_t \rceil \;\leq\; kt.$$

If each test is priced at least 1, the halvings dropped are at most k times the price of the run.

Proof. From $L_{j-1} \leq 2^k L_j$ follows $\lceil \log_2 L_{j-1} \rceil \leq k + \lceil \log_2 L_j \rceil$. Sum over $j = 1, \ldots, t$. □

Example 14.8. On a list of 8 candidates, the single test "is it this one?" can leave 1 candidate: three halvings for one priced step. That test keeps a share of $1/8$, so it is excluded when $k = 1$, and with it the bound for $k = 1$ fails.

Argued in Chapter 12 of the book.

What standing behind a grouping gets you

Let X be a finite set of items. A *grouping* is a map $g : X \to \Lambda$ into a set of labels; two items share a group when they have the same label. A *question* is a map $k : X \to K$. The grouping g is *sound* for k when

$$\forall x, y \in X : \quad g(x) = g(y) \implies k(x) = k(y).$$

A grouping h *cuts* g *smaller* (refines g) when $h(x) = h(y) \Rightarrow g(x) = g(y)$ for all $x, y \in X$.

Theorem 14.9 (Soundness transfers exactly to cuttings)**.** [shown] *Let g, h be groupings of X.*

(a) Soundness depends on a grouping only through which items share a group: if $g(x) = g(y) \Leftrightarrow g'(x) = g'(y)$ for all x, y, then g is sound for k if and only if g' is.

(b) h is sound for every question for which g is sound if and only if h cuts g smaller.

(c) If h does not cut g smaller, there is a question with two values for which g is sound and h is not.

Proof. (a) The definition of soundness mentions g only through the relation $g(x) = g(y)$. (b) If h cuts g smaller and g is sound for k, then $h(x) = h(y)$ gives $g(x) = g(y)$ and so $k(x) = k(y)$. Conversely, g is sound for the question $k = g$, so h is sound for g, which says that h cuts g smaller. (c) Take x, y with $h(x) = h(y)$ and $g(x) \neq g(y)$, and let $k(z) =$ yes when $g(z) = g(x)$ and no otherwise. Each group of g has one value of k, so g is sound for k; h puts x and y together with $k(x) \neq k(y)$. □

Remark 14.10. A grouping that does not cut g smaller can still be sound for a particular question; when k is constant every grouping is sound. Theorem 14.9 says what soundness of g guarantees for every question. A box that records "g was found sound for k" records it for the values of k at the time of the check; a later change to k is not covered.

Argued in Chapter 12 of the book.

15 What insight is

Definition 15.1 (Insight, as priced)**.** [chosen] On a machine with a reading r, an *insight*, in the sense the toll prices, is a step that settles more: a step from s by i with $r(\delta(s, i)) \not\leq r(s)$. On the two-point reading it is a step from no to yes. The definition says nothing about what the settled state shows.

Argued in Chapter 13 of the book.

The *board* is the machine with states $\{\text{NO}, \text{YES}\}$, the two-point reading $r(\text{NO}) = \text{no}$, $r(\text{YES}) = \text{yes}$, and four moves: STAY, WORK and CHECK leave the state where it is, and the stamp STAMP sends both states to YES. Its price list charges $c(\text{STAMP}) = 1$; the other prices are free choices, and $c(\text{STAY}) = c(\text{WORK}) = 0$.

Proposition 15.2 (The price is flat)**.** [shown] *Let c obey the toll for r (Definition 2.1).*

(a) Whether the toll requires $c(i) \geq 1$ at a step from s depends only on the pair $(r(s), r(\delta(s, i)))$. A move that settles more at no state may be priced 0*.*

(b) On the two-point reading, every run from a state reading no *to a state reading* yes *has price at least* 1*.*

(c) *On the board, for every property P of states that holds at* YES*, some run from* NO *ends in a state with P that reads* yes*, at price exactly* 1*. So no property of the end state beyond its reading raises the floor.*

Proof. (a) is the form of the toll. (b) Some step of the run goes from a state reading no to one reading yes; that step settles more and its move is priced at least 1; prices are in $\mathbb{N}$. (c) The run STAMP from NO ends at YES at price 1. □

Argued in Chapter 13 of the book.

Proposition 15.3 (Nothing makes the box wait for a check)**.** [shown] *On the board with any price c*(CHECK) $\in \mathbb{N}$*, the run* STAMP *from* NO *ends at* YES *at price* 1 *with no* CHECK *in it, and the run* CHECK STAMP *from* NO *ends at the same state* YES *at price* $1 + c$(CHECK)*. The two prices differ by c*(CHECK)*, and not at all when c*(CHECK) $= 0$*.*

Proof. CHECK leaves the state where it is, and STAMP sends every state to YES. □

Remark 15.4. A board on which the stamp raises the box only after a passing check is built by adding to the state a mark that a passing CHECK sets and letting STAMP move NO to YES only when the mark is set. That is the established record of Definition 2.4. It is a design choice; nothing in Definition 2.2 requires it. As in Remark 14.2, a paid stamp does not show that its claim is true.

Argued in Chapter 13 of the book.

16 A box over any machine

A *base* is a machine (B, I, β) with a set $B_0 \subseteq B$ of starts and a set $H \subseteq B$ of stopped states. When $|I| = 1$ the base has one next state; otherwise the event below may depend on the move as well as the state.

Definition 16.1 (A box over a base)**.** [chosen] A *box machine over* the base is a machine (X, I, μ) with starts X_0, stopped states H_X, a *box* $\rho : X \to \{\text{no}, \text{yes}\}$, a *tally* $t : X \to \mathbb{N}$, and a map $p : X \to B$ that runs the base faithfully: $p(\mu(x, i)) = \beta(p(x), i)$ for all x, i; $p(X_0) = B_0$; and $x \in H_X \Leftrightarrow p(x) \in H$. The price of a step is $t(\mu(x, i)) - t(x)$. The *five rules* are:

(R1) (answers only to the machine) there is $F : B \times I \times \{\text{no}, \text{yes}\} \to \{\text{no}, \text{yes}\}$ with $\rho(\mu(x, i)) = F(p(x), i, \rho(x))$ for every $x \in X$ and $i \in I$;

(R2) (the tally never goes down) $t(x) \leq t(\mu(x, i))$;

(R3) (the toll) $\rho(x) = \text{no}$ and $\rho(\mu(x, i)) = \text{yes}$ imply $t(\mu(x, i)) - t(x) \geq 1$;

(R4) (stays up) $\rho(x) = \text{yes}$ implies $\rho(\mu(x, i)) = \text{yes}$;

(R5) (something turns it up) some run from a start reaches a state x with $\rho(x) = \text{no}$ and $\rho(\mu(x, i)) = \text{yes}$ for some i.

Rule (R1) quantifies over every state of X, reached or not. The extra components of X beyond the base are arbitrary.

For an *event* $h : B \times I \to \{\text{no}, \text{yes}\}$, the *cup on h* is the rule $(b, r) \mapsto (\beta(b, i),\ r \vee h(b, i))$: the cup turns up the first time the event holds and never turns back.

Theorem 16.2 (A box is a cup)**.** [shown]

(a) If a box machine satisfies (R1) and (R4), then with $h(b,i) = F(b,i,\mathrm{no})$*,*

$$\big(p(\mu(x,i)),\ \rho(\mu(x,i))\big) = \big(\beta(p(x),i),\ \rho(x) \vee h(p(x),i)\big) \qquad \text{for all } x \in X,\ i \in I.$$

Rules (R2), (R3) and (R5) are not used.

(b) Conversely, let $h : B \times I \to \{\mathrm{no},\mathrm{yes}\}$ *be an event that some run of the base from a start reaches: some* b *on such a run and some* i *have* $h(b,i) = \mathrm{yes}$*. On* $X = B \times \{\mathrm{no},\mathrm{yes}\} \times \mathbb{N}$ *let*

$$\mu((b,r,n),i) = \big(\beta(b,i),\ r \vee h(b,i),\ n + [\,r = \mathrm{no} \wedge h(b,i) = \mathrm{yes}\,]\big),$$

with $X_0 = B_0 \times \{\mathrm{no}\} \times \{0\}$*,* $\rho(b,r,n) = r$*,* $t(b,r,n) = n$*,* $p(b,r,n) = b$ *and* $H_X = p^{-1}(H)$*. This machine satisfies (R1) to (R5).*

So the boxes over a base that satisfy the five rules are exactly the cups on events the base reaches, together with a tally and any extra components.

Proof. (a) If $\rho(x) = \mathrm{yes}$, (R4) gives $\rho(\mu(x,i)) = \mathrm{yes} = \rho(x) \vee h(p(x),i)$. If $\rho(x) = \mathrm{no}$, (R1) gives $\rho(\mu(x,i)) = F(p(x),i,\mathrm{no}) = h(p(x),i)$. The first component is faithfulness. (b) (R1) holds with $F(b,i,r) = r \vee h(b,i)$; the tally only adds; a step that turns the cup up has $r = \mathrm{no}$ and $h(b,i) = \mathrm{yes}$ and adds 1; $r \vee h(b,i) = \mathrm{yes}$ when $r = \mathrm{yes}$. For (R5), follow a run of the base from a start to the first point where the event holds; the cup is down there and the next step turns it up. □

Argued in Chapter 14 of the book.

Corollary 16.3 (A turned cup is news, from a clean start)**.** [shown] *Under (R1) and (R4), on a run* $x_0, x_1, \ldots, x_n$ *with* $\rho(x_0) = \mathrm{no}$ *along moves* $i_1, \ldots, i_n$*,* $\rho(x_n) = \mathrm{yes}$ *if and only if* $h(p(x_{j-1}), i_j) = \mathrm{yes}$ *for some* $1 \le j \le n$*. Without* $\rho(x_0) = \mathrm{no}$ *the forward direction fails: a run that starts with the cup up has it up whatever the event does.*

Proof. By Theorem 16.2(a), $\rho(x_n) = \rho(x_0) \vee \bigvee_j h(p(x_{j-1}), i_j)$. □

Argued in Chapter 14 of the book.

Example 16.4 (Both rules do work)**.** [shown] Let the base be $(\mathbb{N}, \{*\}, n \mapsto n+1)$.

(a) (The toggle) On $X = \mathbb{N} \times \{\mathrm{no},\mathrm{yes}\}$ let $\mu(n,r) = (n+1, \neg r)$. Rule (R1) holds with $F(b,r) = \neg r$; (R4) fails; and the box is not the cup on any event, since a cup that is up stays up.

(b) (The timestamp) On $X = \mathbb{N} \times \mathbb{N} \times \{\mathrm{no},\mathrm{yes}\}$, with a hidden clock k, let $\mu(n,k,r) = (n+1, k+1, r \vee [k = 5])$. Rule (R4) holds. Rule (R1) fails: the states $(n,5,\mathrm{no})$ and $(n,0,\mathrm{no})$ have the same base state and the same box, and their next boxes differ.

(c) (The timestamp started in step) With the only start $(0,0,\mathrm{no})$, every reached state has $k = n$, and on reached states the box evolves as the cup on the event $[n = 5]$. Rule (R1) still fails, because it quantifies over every state.

Argued in Chapter 14 of the book.

Lemma 16.5 (Finite chairs)**.** [shown] *Let S be finite, let $r : S \to (R, \leq)$ satisfy $r(s) \leq r(\delta(s,i))$ for all s, i, and let the move i settle more at some s. Then there are $x \neq y$ in S with $\delta(x,i) = \delta(y,i)$.*

Proof. Let $T = \{t \in S : r(\delta(s,i)) \leq r(t)\}$. Then $\delta(s,i) \in T$, and $t \in T$ implies $\delta(t,i) \in T$ by transitivity, since $r(t) \leq r(\delta(t,i))$. Also $s \notin T$, because s settles more. So the move i maps the $|T|+1$ states of $T \cup \{s\}$ into the $|T|$ states of T, and two of them land together. □

Proposition 16.6 (A base that never merges)**.** [shown] *Let the base have one next state $f : B \to B$ with f one-to-one, and let $h : B \to \{\text{no}, \text{yes}\}$ be an event. On $X = B \times \{\text{no}, \text{yes}\} \times \{\text{no}, \text{yes}\}^*$, where $\{\text{no}, \text{yes}\}^*$ is the set of finite lists, let $\mu(b, r, \sigma) = (f(b),\ r \vee h(b),\ r\sigma)$: the strip σ keeps every earlier position of the cup, newest first. Then μ is one-to-one, the box evolves as the cup on h, and the strip grows by one entry each step, so no run visits a state twice. If instead X is finite and the box stays up, a move that turns the box up makes two arrows meet (Lemma 16.5).*

Proof. If $\mu(b, r, \sigma) = \mu(b', r', \sigma')$, then $r\sigma = r'\sigma'$ gives $r = r'$ and $\sigma = \sigma'$, and $f(b) = f(b')$ gives $b = b'$. The length of the strip is the number of steps taken. □

Argued in Chapter 14 of the book.

Proposition 16.7 (Two choices nobody forces)**.** [shown]

(a) (The price list) Let a box machine satisfy (R1) to (R5). On $X \times \mathbb{N}$ let $\mu'((x,m),i) = (\mu(x,i),\ m + t(\mu(x,i)) - t(x) + 1)$, with box $\rho(x)$, tally m, map $p(x)$, and starts $X_0 \times \{0\}$. This machine satisfies (R1) to (R5), its box evolves exactly as before, and every one of its steps is priced one more than the same step of the original.

(b) (The event) Let h_1, h_2 be events each reached by some run of the base from a start. Both cups of Theorem 16.2(b) satisfy (R1) to (R5). At any b, i with $h_1(b,i) = \text{yes}$ and $h_2(b,i) = \text{no}$, from a down cup the first cup goes up and the second stays down.

Proof. (a) The box and p are unchanged, so (R1), (R4) and (R5) carry over; the new increment is the old one plus 1, which is at least 1, giving (R2) and (R3). (b) Theorem 16.2(b) and the formula for the cup. □

Example 16.8 (A cup and a die)**.** Over a list of die rolls read one entry per step, the cup on "this entry is a six" and the cup on "this entry is a one" both satisfy the five rules. If a one is rolled first, the two cups disagree until a six appears.

Argued in Chapter 14 of the book.

A *random box* is the box alone, $\{\text{no}, \text{yes}\}$, with a chance of each next position: from yes the next position is yes with chance 1; from no it is yes with chance q and no with chance $1 - q$. The step from no to yes is priced 1 and every other step 0.

Proposition 16.9 (Prices do not fix chances)**.** [shown] *The random boxes with $q = 1/2$ and with $q = 2/3$ have the same possible steps, the same prices, obey the toll, and keep the box up once up, and they differ in their chances. Neither is a cup on any event: from the one state* no *both next positions are possible, so the next position is not a function of the present state.*

Argued in Chapter 14 of the book.

Proposition 16.10 (One change from no to yes)**.** [cited] *(The bottom rung of the hierarchy of guessing with a bounded number of changes, after Putnam, Gold and Ershov.) A set $A \subseteq \mathbb{N}$ is listed by some program if and only if there is a computable $g : \mathbb{N} \times \mathbb{N} \to \{\text{no}, \text{yes}\}$ with $g(x, 0) = \text{no}$, $g(x, s) \leq g(x, s+1)$ for all x, s, and $x \in A \Leftrightarrow \exists s\ g(x, s) = \text{yes}$.*

Proof. If a program lists A, let $g(x, s) = \text{yes}$ when x appears among the first s outputs. Conversely, run $g(x, s)$ for all pairs (x, s) in a fixed order and output x the first time $g(x, s) = \text{yes}$. □

Remark 16.11. On this rung a guess yes is final when it appears and a guess no never is. A single cup can record that an event happened and cannot record that it did not.

Argued in Chapter 14 of the book.

17 The machine with the record dropped

Let (S, I, δ) carry a reading $r : S \to (R, \leq)$ that never goes back: $r(s) \leq r(\delta(s, i))$. A step *turns up a cup* when it settles more; for a reading into a chain or a checklist these are the steps that turn up some cup of the row. Let $\pi : S \to C$ be the *covering map* onto the *uncovered machine* (C, I, γ), with

$$\pi(\delta(s, i)) = \gamma(\pi(s), i) \qquad \text{for all } s \in S,\ i \in I.$$

This equation is the condition that nothing in how the rest of the machine moves depends on the cups or the tally. The pile over $c \in C$ is $\pi^{-1}(c)$. The *landing pile* of a move i at $y \in S$ is $\{x \in S : \delta(x, i) = y\}$. The uncovered machine *forgets nothing* at i when $\gamma(\cdot, i)$ is one-to-one.

Theorem 17.1 (What a paid move crushes)**.** [shown]

(a) If the uncovered machine forgets nothing at i, every landing pile of i lies inside one pile: $\delta(x, i) = \delta(y, i)$ implies $\pi(x) = \pi(y)$.

(b) If moreover S is finite and i settles more at some state, there are $x \neq y$ with $\pi(x) = \pi(y)$ and $\delta(x, i) = \delta(y, i)$: the move crushes two states of one pile into one.

(c) Without the condition in (a) the merge can join two piles: on the board of Section 15 with $C = S$ and π the identity, STAMP *sends* NO *and* YES *to* YES*, and* NO, YES *lie in different piles.*

Proof. (a) $\gamma(\pi(x), i) = \pi(\delta(x, i)) = \pi(\delta(y, i)) = \gamma(\pi(y), i)$, and $\gamma(\cdot, i)$ is one-to-one. (b) Lemma 16.5 gives $x \neq y$ with $\delta(x, i) = \delta(y, i)$; apply (a). (c) is direct. □

Argued in Chapter 15 of the book.

Proposition 17.2 (The halving rule is a bound on landing piles)**.** [shown] *For a move i and $c(i) \in \mathbb{N}$, on a machine of any size, the following are equivalent:*

(i) for every finite $D \subseteq S$, $|D| \leq 2^{c(i)} \cdot |\{\delta(x, i) : x \in D\}|$ (the halving rule at i);

(ii) every landing pile of i has at most $2^{c(i)}$ elements.

Under (ii), $c(i) \geq \log_2$ of the size of every landing pile of i. With Theorem 17.1(a), the states a landing pile collects all lie in one pile, so $c(i)$ is at least $\log_2$ of the number of states of one pile that i lands together.

Proof. (i)⇒(ii): a finite subset D of one landing pile has one image, so $|D| \le 2^{c(i)}$. (ii)⇒(i): split D by its image; each part lies in one landing pile and has at most $2^{c(i)}$ elements. □

Argued in Chapter 15 of the book.

A clerk prices each step from the window: a price of the form $c(s,i) = q(\pi(s), \pi(\delta(s,i)))$. It *never misses* when $c(s,i) \ge 1$ at every step that turns up a cup, and it is *exact* when it also has $c(s,i) = 0$ at every step that turns up no cup. A *clash* is a pair of steps from s by i and from s' by i' with $\pi(s) = \pi(s')$ and $\pi(\delta(s,i)) = \pi(\delta(s',i'))$, where the first turns up a cup and the second does not.

Proposition 17.3 (A clash forbids an exact price)**.** [shown] *If the window has a clash, no price from the window is exact.*

Proof. A price that never misses charges $q(\pi(s), \pi(\delta(s,i))) \ge 1$, and the same value is charged at the second step of the clash, which turns up no cup. □

Argued in Chapter 15 of the book.

Proposition 17.4 (Showing the cups move)**.** [shown] *Let $S = \mathbb{N}$ with the usual order as the reading (the cup $j \ge 1$ is "at least j"), and two moves:* add$(n) = n + 1$ *and* stay$(n) = n$.

(a) The window $\pi(n) = [n \ge 1]$, which shows only the first cup, has a clash: from $n = 1$, add *turns up a cup and* stay *does not, and both are seen as* (yes, yes).

(b) The window $\pi(n) = n \bmod 2$ has no clash. The price $q(a,b) = [a \ne b]$, one exactly when the parity changed, is exact. No constant price is exact, and no function of $n \bmod 2$ returns n.

Proof. (a) is direct. (b) add turns up a cup at every n and changes the parity; stay turns up none and keeps it. A constant price 0 misses add and a constant price ≥ 1 charges stay. The states 0 and 2 have the same parity. □

Remark 17.5. When the window has a clash, the least overcharge and a price that reaches it are as in Chapter 7 of the book, with a row of cups in place of one box. If the window has no clash an exact price exists.

Argued in Chapter 15 of the book.

18 Reading the box

A *shortcut* is a move that answers a question about the data by a glance at the state instead of working it out the long way. It is *sound* when it gives the answer the long way gives.

Definition 18.1 (The edition machine)**.** [chosen] Fix a set V of values and a property $P \subseteq V$ (the *passing* values). A state is (x, e, U) with $x \in V$ the data, $e \in \mathbb{N}$ the *edition*, and $U \subseteq \mathbb{N}$ the editions whose box is up. The moves are

$$\text{write}_a(x,e,U) = (a, e+1, U), \qquad \text{ask}(x,e,U) = (x,e,U),$$

$$\text{check}(x,e,U) = \begin{cases} (x, e, U \cup \{e\}) & x \in P, \\ (x,e,U) & x \notin P. \end{cases}$$

The shortcut *fires* at (x,e,U) when $e \in U$; it answers as if $x \in P$. The clean start is $(x_0, 0, \varnothing)$. The boxes never come down, the box for e goes up only at a passing check on edition e, and a write moves to an edition never used before: this is the second door.

Theorem 18.2 (Fresh editions are never wrong)**.** [shown] *For every list of moves τ and every $x_0 \in V$, if the shortcut fires at* $\mathrm{run}(\tau, (x_0, 0, \varnothing)) = (x, e, U)$*, then $x \in P$.*

Proof. The property "every $e' \in U$ has $e' \le e$, and $e \in U$ implies $x \in P$" holds at the clean start and is kept by every move. A write sets e to $e+1$, which exceeds every element of U, so $e+1 \notin U$. A passing check adds e and has $x \in P$; a failing check and ask change nothing. □

Argued in Chapter 16 of the book.

Proposition 18.3 (Editions that wrap around)**.** [shown] *(The first door.) Replace $e+1$ by $(e+1) \bmod 2$ in* write_a*, and let $V = \mathbb{N}$ and P the even numbers. From $(0, 0, \varnothing)$ the four moves* check*,* write_1*,* write_3*,* ask *reach $(3, 0, \{0\})$, where the shortcut fires and $3 \notin P$: a wrong answer.*

Proof. $(0, 0, \varnothing) \mapsto (0, 0, \{0\}) \mapsto (1, 1, \{0\}) \mapsto (3, 0, \{0\}) \mapsto (3, 0, \{0\})$. □

Argued in Chapter 16 of the book.

Proposition 18.4 (A box that comes down merges twice)**.** [shown] *(The third door.) Let $V = \{0, 1\}$ with $P = \{1\}$, states $(x, b) \in V \times \{\mathrm{no}, \mathrm{yes}\}$, the passing check* $\mathrm{check}(x, b) = (x, \mathrm{yes})$ *if $x = 1$ and (x, b) otherwise, and the write that flips the data and knocks the box down,* $\mathrm{write}(x, b) = (1 - x, \mathrm{no})$*. Then $x \mapsto 1 - x$ is one-to-one, and still*

$$\mathrm{write}(0, \mathrm{no}) = \mathrm{write}(0, \mathrm{yes}), \qquad \mathrm{check}(1, \mathrm{no}) = \mathrm{check}(1, \mathrm{yes}) :$$

two arrows meet at the wipe and at the check. The box comes down, so it is not a reading that never goes back.

Argued in Chapter 16 of the book.

The four doors

The four doors for keeping a shortcut honest after writes are: editions that wrap around (the first door); editions never reused, which on a finite machine is a budget (the second); one box knocked down by every write (the third); and a box tied to the value checked instead of the edition (the fourth). The *guard* is the test that decides whether the shortcut may run; when it says yes the shortcut is *unlocked*. The guard reads a *memory* of E boxes, $\{\mathrm{no}, \mathrm{yes}\}^E$, ordered coordinatewise; $|m|$ is the number of boxes up in m. A *round* is: write a passing value, check it, ask; then write a failing value, and ask again.

Theorem 18.5 (Four doors)**.** [shown] *Assume:*

(C1) the guard's glance cannot tell a passing value from a failing one: whether the shortcut is unlocked is $g(m)$ for a fixed $g : \{\mathrm{no}, \mathrm{yes}\}^E \to \{\mathrm{no}, \mathrm{yes}\}$ of the memory m alone;

(C2) the memory is made only of boxes that go up and never come down: each move takes the memory m to some $m' \ge m$.

Run n rounds from a memory b_0. Let a_j be the memory after the check of round j and b_j the memory after its failing write, so $b_0 \le a_1 \le b_1 \le a_2 \le \cdots \le b_n$. Round j unlocks *when $g(a_j) =$* yes*, and* answers wrong *when $g(b_j) =$* yes*, since the shortcut then runs on a failing value. If no round answers wrong, the number u of rounds that unlock satisfies*

$$2u \;\le\; E + 1, \qquad \textit{that is,} \qquad u \le \frac{E+1}{2}.$$

So under (C1) and (C2), on any number of rounds, either the shortcut answers wrong (the first door), or it unlocks on at most $(E+1)/2$ rounds and takes the long way on every round after those (a budget: the second door).

Proof. Suppose $g(b_j) = \text{no}$ for $1 \le j \le n$. If round j unlocks, then $a_j \neq b_j$ and $a_j \le b_j$, so $|b_j| \ge |a_j| + 1$. If also $g(b_{j-1}) = \text{no}$, then $b_{j-1} \neq a_j$ and $|a_j| \ge |b_{j-1}| + 1$; this holds for every $j \ge 2$, and for $j = 1$ unless $g(b_0) = \text{yes}$. The intervals from b_{j-1} to a_j and from a_j to b_j for the unlocking rounds are disjoint pieces of the chain $b_0 \le a_1 \le \cdots \le b_n$, along which $|\cdot|$ never decreases. So

$$E \;\ge\; |b_n| - |b_0| \;\ge\; 2u - [\,g(b_0) = \text{yes}\,] \;\ge\; 2u - 1. \qquad \square$$

Remark 18.6 (Outside the two conditions are the other two doors). If (C2) fails, a box that a write takes down unlocks every round with no wrong answer: with $E = 1$, the check setting the box up, the failing write setting it down, and g the identity, every round unlocks. That is the third door, and by Proposition 18.4 it merges at the wipe and at the check. If (C1) fails, the glance reads enough of the value to tell passing from failing; either it repeats the check, which is no shortcut, or it ties a box to the value, which is the fourth door, priced in room by Proposition 18.7.

Argued in Chapter 16 of the book.

Proposition 18.7 (Keeping k values unlocked takes 2^k memory states)**.** [shown] *Let K be a set of k passing values, M a set of memory states, and $g : M \times V \to \{\text{no}, \text{yes}\}$ the guard. Suppose that for every set $T \subseteq K$ of checked values there is a memory state $m_T \in M$ that unlocks exactly the checked values: $g(m_T, v) = \text{yes} \Leftrightarrow v \in T$ for all $v \in K$. Then $T \mapsto m_T$ is one-to-one, so $|M| \ge 2^k$. One box per value, $M = \{\text{no}, \text{yes}\}^K$, meets this with $|M| = 2^k$.*

Proof. If $T \neq T'$, some $v \in K$ lies in exactly one of them, and $g(m_T, v) \neq g(m_{T'}, v)$, so $m_T \neq m_{T'}$. There are 2^k sets T. $\square$

Argued in Chapter 16 of the book.

Remark 18.8. [not claimed] It is not claimed that reading the box computes something new. Every answer a sound shortcut gives, the long way gives too. What reading the box saves is steps, the way a cache does.

Argued in Chapter 16 of the book.

19 What a machine can't decide about itself

Let $\mathcal{P}$ be a set of programs with an equivalence $\approx$, "behaves alike". Let $\mathcal{R}$ be a set of maps $\mathcal{P} \to \mathcal{P}$, the transformations the machine can run. The machine has *Kleene's trick* for $\mathcal{R}$ when for every $\phi \in \mathcal{R}$ there is $p \in \mathcal{P}$ with $p \approx \phi(p)$. A question $A \subseteq \mathcal{P}$ ("has a shortcut") is *about behaviour* when $p \approx q$ implies $p \in A \Leftrightarrow q \in A$, and it is *not trivial* when there are $p_{\text{yes}} \in A$ and $p_{\text{no}} \notin A$. For $d : \mathcal{P} \to \{\text{no}, \text{yes}\}$, the *flip* of d is $\phi_d(p) = p_{\text{no}}$ if $d(p) = \text{yes}$ and $\phi_d(p) = p_{\text{yes}}$ if $d(p) = \text{no}$.

Theorem 19.1 (No inside autopilot)**.** [shown] *Let the machine have Kleene's trick for $\mathcal{R}$, and let A be about behaviour and not trivial. There is no $d : \mathcal{P} \to \{\text{no}, \text{yes}\}$ with $\phi_d \in \mathcal{R}$ and*

$$\forall p \in \mathcal{P}: \quad d(p) = \text{yes} \iff p \in A.$$

Proof. Suppose d is one. Kleene's trick gives p with $p \approx \phi_d(p)$. If $d(p) =$ yes, then $p \in A$ and $p \approx p_{\text{no}}$, so $p_{\text{no}} \in A$, which is false. If $d(p) =$ no, then $p \notin A$ and $p \approx p_{\text{yes}}$, so $p_{\text{yes}} \notin A$, which is false. □

Remark 19.2. The argument uses only Kleene's trick, that the flip can be run, the two examples p_{yes} and p_{no}, and that A is about behaviour. It does not use a box, a tally or the toll, and it does not see soundness: whether a particular question about shortcuts is about behaviour and not trivial is a separate fact about that machine. It rules out an inside procedure, one whose flip the machine can run, that is right on every program. It says nothing against a procedure right on some programs, one allowed to decline, one that uses outside evidence, or a judge outside the machine. It is Rice's form of the diagonal, with its own conditions, and says nothing about how hard the question is compared with stopping.

Argued in Chapter 17 of the book.

Remark 19.3. [not claimed] It is not claimed that undecidable means unknowable. For each single program p, one of the two constant procedures "say yes" and "say no" is right on p. Theorem 19.1 rules out one inside procedure that is right on every program.

Argued in Chapter 17 of the book.

Proposition 19.4 (An established box does not rest on itself)**.** [shown] *Let the box be established for a claim κ (Definition 2.4) and never come down. On any run from a clean start, the box goes up at most once, and the passing check of κ that precedes the raise is made from a state where the box is down and lands in a state where the box is down. Hence if a check of κ passes only at states where the box is up, no run from a clean start raises the box.*

Proof. Let the run be $s_0, s_1, \ldots$ with the box down at s_0, and let the raise be the step from s_j to s_{j+1}. Since the box never comes down, it is down at $s_0, \ldots, s_j$ and up from s_{j+1} on, so there is no second raise. The passing check is a step from s_k to s_{k+1} with $k < j$, and $k + 1 \leq j$. □

Argued in Chapter 17 of the book.

Remark 19.5. Two limits stay apart. Theorem 19.1 is about what no inside procedure can decide. The window of Chapter 7 of the book is about what no outside watcher can see: two states with different boxes and the same view cannot be told apart through that view, on small finite machines with nothing to compute.

Argued in Chapter 17 of the book.

20 Five real systems

Throughout Sections 20 to 24, a two-point reading is called a *box*. The box is *up* at s when $r(s) =$ yes and *down* when $r(s) =$ no. A step from s by i *raises* the box when $r(s) =$ no and $r(\delta(s, i)) =$ yes. For a list of moves $\tau = i_1 \cdots i_n$ and a start s, raises(τ, s) is the number of k such that the step from s_{k-1} by i_k raises the box. A machine with one next state f *makes two arrows meet* on a set D when $f(x) = f(y)$ for some $x \neq y$ in D.

Definition 20.1 (Fee, bond, budget)**.** [chosen] A system *prices* an event when it attaches to it one of three costs. A *fee*: something taken at the event itself, whatever happens afterwards. A *bond*: something put at risk by the event and lost only if some stated later event happens. A *budget*: a limited allowance the event draws down, so that only finitely many such events can happen. A cost of undoing that rests only on a computation nobody knows how to reverse (a *wall*) is not a price, and neither is work a system needs and never charges for.

Argued in Chapter 18 of the book.

Proof-of-stake finality

Definition 20.2 (Checkpoints, votes, finality)**.** [chosen] Let $\mathcal{T}$ be a finite rooted tree of *checkpoints* with root g (the genesis block), and let $h(x)$ be the depth of x (its *round*). Write $x \preceq y$ when x is an ancestor of y or $x = y$. Two checkpoints *conflict* when neither is $\preceq$ the other. Let V be a finite set of validators with stakes $w : V \to (0, \infty)$, $W = \sum_{v \in V} w(v)$, and $w(U) = \sum_{v \in U} w(v)$. A *vote* of a validator is a pair (s, t) of checkpoints with $s \preceq t$, $s \neq t$; it is a *link* from s to t. A link is a *supermajority link* when the validators who voted it have stake at least $2W/3$. The *justified* checkpoints form the least set containing g such that t is justified whenever s is justified and $s \to t$ is a supermajority link. A checkpoint s is *final* when it is justified and there is a supermajority link from s to a child c of s. A validator is *slashable* when it cast two votes (s_1, t_1), (s_2, t_2) with either $t_1 \neq t_2$ and $h(t_1) = h(t_2)$ (a double vote), or $h(s_1) < h(s_2) < h(t_2) < h(t_1)$ (a surround vote).

Theorem 20.3 (Accountable safety)**.** [shown on the copy] *In the setting of Definition 20.2, if two conflicting checkpoints are both final, then the slashable validators have stake at least $W/3$.*

Proof. Two sets $U_1, U_2 \subseteq V$ with $w(U_1), w(U_2) \geq 2W/3$ satisfy $w(U_1 \cap U_2) \geq w(U_1) + w(U_2) - W \geq W/3$. Every justified $t \neq g$ has a supermajority link into it from a justified checkpoint, and $g \preceq x$ for every x, so neither final checkpoint is g. Let f_1, f_2 be the two, with $v = h(f_1) \leq h(f_2)$, and let $f_1 \to c$ be a finalizing link, $h(c) = v + 1$. If $h(f_2) = v$, the links into f_1 and f_2 land in round v on different checkpoints, and their voter sets share stake at least $W/3$, all of it double votes. If $h(f_2) > v$, take a chain of supermajority links $g = j_0 \to j_1 \to \cdots \to j_n = f_2$ with every j_l justified. Every $j_l \preceq f_2$, so no j_l is f_1 or below f_1 (else $f_1 \preceq f_2$). Rounds increase along the chain from $0 \leq v$ to $h(f_2) > v$, so some l has $h(j_l) \leq v < h(j_{l+1})$. Let U vote $j_l \to j_{l+1}$ and U' vote $f_1 \to c$. If $h(j_{l+1}) = v + 1$, then $j_{l+1} \neq c$ because $f_1 \preceq c$, and $U \cap U'$ made double votes. If $h(j_l) = v$, then $j_l \neq f_1$ are both justified in round $v \geq 1$, and the voters shared by their two justifying links made double votes. Otherwise $h(j_l) < v$ and $h(j_{l+1}) > v + 1$, so $h(j_l) < h(f_1) < h(c) < h(j_{l+1})$ and $U \cap U'$ made surround votes. In each case the slashable stake is at least $W/3$. □

Argued in Chapter 18 of the book.

Example 20.4 (Forks and the overlap)**.** [shown] (a) Three validators of stake 1; g in round 0, two children L_1, R_1 in round 1, a child L_2 of L_1 and R_2 of R_1 in round 2. Let A, B vote $g \to L_1$ and $L_1 \to L_2$, and B, C vote $g \to R_1$ and $R_1 \to R_2$. Then L_1 and R_1 are both final and conflict; B is slashable (double vote in round 1); A and C are not. The slashable stake is $1 = W/3$. (b) With one validator, the single vote $g \to c$ makes g final and no validator is slashable. (c) Replace the threshold $2W/3$ by $W/2$, with two validators of stake 1, A voting the left links and B the right links of (a). Then L_1 and R_1 are both final and no validator is slashable. The theorem rests on the overlap of two sets of stake $2W/3$.

Argued in Chapter 18 of the book.

Gas

Definition 20.5 (One storage slot)**.** [chosen] A slot holds a value in $\mathbb{N}$ and starts at 0. A *store* of x into a slot holding y costs 20000 if $y = 0 \neq x$ and 100 otherwise, and credits a refund of

19900 if $y \neq 0 = x$. If the slot is *cold*, the first store costs a further 2100. The *net* after a list of stores is the sum of the costs minus the sum of the refunds.

Proposition 20.6 (The floor on a write left standing)**.** [shown on the copy] *After any list of stores, the net is at least* 0 *if the slot holds* 0 *and at least* 20000 *if it does not. From a cold slot, the net when the slot ends nonzero is at least* 22100*.*

Proof. Ignore the cold charge. The two bounds hold at the start (net 0, slot 0). A store from 0 to 0 adds 100; from 0 to $x \neq 0$ adds 20000; from $y \neq 0$ to $x \neq 0$ adds 100; from $y \neq 0$ to 0 adds $100 - 19900 = -19800$, taking a net of at least 20000 to at least $200 \geq 0$. Each store preserves the bound that fits where it lands. The cold charge is added once and changes nothing else. □

Argued in Chapter 18 of the book.

Example 20.7 (Set and clear)**.** [shown on the copy] From a cold slot, storing 1 and then 0 costs $22100 + 100 = 22200$ and credits 19900: net 2300. Under a cap that limits the refund to one fifth of the gas a transaction uses, a transaction made of a base charge of 21000 and these two stores uses 43200, receives $\min(19900, 8640) = 8640$ and pays 34560, which is 13560 above the base. Applied to the two stores alone, the cap gives $22200 - \min(19900, 4440) = 17760$.

Argued in Chapter 18 of the book.

Proposition 20.8 (An exact schedule)**.** [shown] *Let* (S, I, δ) *carry a box and a price* $c : S \times I \to \mathbb{N}$ *that may depend on the state a step starts from. For a run, let* $c(\tau, s)$ *be the sum of the prices of its steps. The following are equivalent.*

(a) (floor and ceiling) $\text{raises}(\tau, s) \leq c(\tau, s) \leq \text{raises}(\tau, s)$ *for every* $s \in S$ *and every* τ*;*

(b) $c(s, i) \geq 1$ *whenever the step from* s *by* i *raises the box, and* $c(\tau, s) \leq \text{raises}(\tau, s)$ *for every* s *and* τ*;*

(c) $c(s, i) = 1$ *when the step from* s *by* i *raises the box, and* $c(s, i) = 0$ *otherwise.*

Proof. (c) gives $c(\tau, s) = \text{raises}(\tau, s)$ by summing, hence (a). (a) applied to one-step runs gives (c), and (c) gives (b). From (b), a one-step run gives $c(s, i) \leq 1$ at a raise and $c(s, i) \leq 0$ elsewhere; with the floor at raises, (c) follows. □

Argued in Chapter 18 of the book.

Attestation

Proposition 20.9 (What a quote cannot say)**.** [shown] *Let a machine's state be a pair* (ρ, σ) *of register values* ρ *and the rest* σ*, and let a quote be any function* $Q(\rho, n)$ *of the registers and a challenge* n*. If* $\rho = \rho'$*, then* $R(Q(\rho, n)) = R(Q(\rho', n))$ *for every reader* R *and every* n*. In particular, if* (ρ, σ) *has a flag up and* (ρ, σ') *has it down, no reader of the quote returns the flag on both. Separately, for any signing map* sign*, the check* verify $\equiv$ yes *satisfies* $\text{verify}(pk, m, \text{sign}(sk, m)) = \text{yes}$ *for all* pk, sk, m *and accepts every signature.*

Proof. Equal arguments give equal values. A reader returning the flag would give two values on one quote. The last claim holds by inspection. □

Argued in Chapter 18 of the book.

Certificate transparency

Example 20.10 (Seven cards)**.** [shown on the copy] Work with brackets: a leaf is $\langle d \rangle$ and a node is $[x\,y]$, two terms being equal only when they are written the same. For a list D of $n \geq 1$ items, $\mathrm{MTH}(D) = \langle d_0 \rangle$ if $n = 1$, and otherwise $\mathrm{MTH}(D) = [\mathrm{MTH}(D_{<k})\ \mathrm{MTH}(D_{\geq k})]$ with k the largest power of two below n. For $D = (d_0, \ldots, d_6)$, with $a = [\langle d_0 \rangle \langle d_1 \rangle]$, $b = [\langle d_2 \rangle \langle d_3 \rangle]$, $c = [\langle d_4 \rangle \langle d_5 \rangle]$, $L = [a\,b]$ and $R = [c\,\langle d_6 \rangle]$, the root is $\mathrm{MTH}(D) = [L\,R]$. Joining in order $\langle d_2 \rangle$ on the left, a on the left and R on the right rebuilds the root from $\langle d_3 \rangle$. The proofs for the other leaves are: d_0: $\langle d_1 \rangle, b, R$, all on the right; d_4: $\langle d_5 \rangle, \langle d_6 \rangle$ on the right, then L on the left; d_6: c, then L, both on the left. The tree of the first four cards has root $L = \mathrm{MTH}(d_0, \ldots, d_3)$, and $[L\,R]$ is the root of seven: the old tree is the left half of the new one. Since $[\langle d_3 \rangle \langle d_2 \rangle] \neq b$, a proof must carry the side of each join. That distinct trees have distinct roots holds for brackets; for a real log it is a further assumption about the hash function.

Argued in Chapter 18 of the book.

The fourth currency

Proposition 20.11 (A permanent write merges on its own yes side)**.** [shown] *Let S be finite, $f : S \to S$, and r a box with $r(s) = \mathrm{yes} \Rightarrow r(f(s)) = \mathrm{yes}$. Let Y be the set of states where the box is up and $s_0 \notin Y$ with $f(s_0) \in Y$. Then there are $x \neq y$ in $Y \cup \{s_0\}$ with $f(x) = f(y) \in Y$.*

Proof. f maps the $|Y| + 1$ states of $Y \cup \{s_0\}$ into Y; two of them land together. □

Argued in Chapter 18 of the book.

Proposition 20.12 (The count)**.** [shown on the copies] *Classified by Definition 20.1, on the pieces copied in the book: Casper attaches a bond to the write and takes it only on contradicting votes; gas attaches a fee to the write and a budget (the gas limit of each block); the quoted registers of the security chip, certificate transparency and proof-carrying code attach none of the three. Of the fifteen fee, bond and budget entries, three are filled, one of each kind, and exactly one system takes a fee at the moment of the write.*

Argued in Chapter 18 of the book.

Remark 20.13. [not claimed] A copy of one piece of a system is not claimed to be the running system. Each statement above is about the piece as stated.

Remark 20.14. [not claimed] It is not claimed that a toll is a better design than a wall.

21 Which box is worth keeping

Definition 21.1 (Pointer)**.** [chosen] Fix a machine, its set Reach of reachable states, and a set $\mathcal{E}$ of *events*, each a set $E \subseteq S$ of states (the event holds at s when $s \in E$). Fix a test $\mathcal{P} \subseteq \mathcal{E}$ (the events copied and relied on by parties who did not make them), chosen without reference to the established record. An event E is a *pointer* for $\mathcal{P}$ when $E \in \mathcal{P}$ and $E \cap \mathrm{Reach} \subseteq E'$ for every $E' \in \mathcal{P}$.

Proposition 21.2 (A pointer is unique on reachable states)**.** [shown] *If E_1 and E_2 are pointers for $\mathcal{P}$, then $E_1 \cap \mathrm{Reach} = E_2 \cap \mathrm{Reach}$. A pointer need not exist.*

Proof. $E_1 \cap \mathrm{Reach} \subseteq E_2$ and $E_2 \cap \mathrm{Reach} \subseteq E_1$. For non-existence, take $\mathcal{P} = \{E, E'\}$ with $E \cap \mathrm{Reach}$ and $E' \cap \mathrm{Reach}$ incomparable. □

Argued in Chapter 19 of the book.

Definition 21.3 (Separate carriers)**.** [chosen] Readers $j \in J$ have *separate carriers* when the world's state space is a product $\prod_{j \in J} W_j$ and the view of reader j depends only on the coordinate w_j. The coordinates may be set independently; the states a machine reaches may still tie them together.

Thesis 21.4 (The pointer thesis)**.** [thesis] For record-keeping as people run it, modelled faithfully, with a test of reliance chosen for its own reasons and readers with separate carriers: (1) the established record (Definition 2.4) is the pointer, so every event that is copied and relied on by parties who did not make it holds wherever the established record does, on the reachable states; and (2) every record the system prices is copied and relied on by parties who did not make it.

Argued in Chapter 19 of the book.

Example 21.5 (A rival event that is copied and relied on)**.** [shown] Let a block's base fee be b, its target use g^* (half the block limit) and its use g. The next base fee is $b + b(g - g^*)/(8g^*)$, rounded down. A full block ($g = 2g^*$) takes 800 to 900; an empty block takes 800 to 700. Every copy of the chain reads the last block's use and acts on it.

Argued in Chapter 19 of the book.

Definition 21.6 (Three states)**.** [chosen] $S = \{\mathsf{blank}, \mathsf{once}, \mathsf{twice}\}$, box up at once and twice. Two moves: flip sends $\mathsf{blank} \mapsto \mathsf{once}$ and fixes once, twice; smooth sends $\mathsf{once} \mapsto \mathsf{twice}$ and fixes blank, twice. Prices $c : S \times I \to \mathbb{N}$ depend on the state a step starts from. The requirements that never name the reading are: (R1) $\delta(s, i) = s \Rightarrow c(s, i) = 0$; (R2) if $\delta(\cdot, i)$ is one-to-one, then $c(s, i) = 0$ for all s; (R3) $c(\varphi(s), i) = c(s, i)$ for every bijection φ of S with $\varphi(\delta(s, i)) = \delta(\varphi(s), i)$ for all s, i; (R4) the price of a run is the sum of the prices of its steps; (R5) the price of a step is a function of the state it starts from and the move; (R6) $c(\mathsf{blank}, \mathsf{flip}) = 1$. For $n \in \mathbb{N}$ let $c_n(\mathsf{blank}, \mathsf{flip}) = 1$, $c_n(\mathsf{once}, \mathsf{smooth}) = n$, and $c_n = 0$ on the other four steps.

Proposition 21.7 (The list does not choose)**.** [shown] *A price $c : S \times I \to \mathbb{N}$ on the three states meets (R1) to (R6) if and only if $c = c_n$ for some $n \in \mathbb{N}$. Both moves make two arrows meet. c_0 is the toll (one on the raise, nothing else), c_1 is a price on merging, and they differ at* $(\mathsf{once}, \mathsf{smooth})$.

Proof. (R4) and (R5) hold of every such c. flip sends $\mathsf{blank}, \mathsf{once}$ to once and smooth sends $\mathsf{once}, \mathsf{twice}$ to twice, so (R2) asks nothing. A bijection φ commuting with both moves fixes their only common fixed point twice, maps the fixed points $\{\mathsf{once}, \mathsf{twice}\}$ of flip to themselves, so fixes once, and then blank; so (R3) asks nothing. (R1) sets the steps $(\mathsf{once}, \mathsf{flip})$, $(\mathsf{twice}, \mathsf{flip})$, $(\mathsf{blank}, \mathsf{smooth})$, $(\mathsf{twice}, \mathsf{smooth})$ to 0, (R6) sets $(\mathsf{blank}, \mathsf{flip})$ to 1, and nothing constrains $(\mathsf{once}, \mathsf{smooth})$. □

Argued in Chapter 19 of the book.

Proposition 21.8 (The ceiling that names the reading chooses)**.** [shown] *On any machine with a box and $c : S \times I \to \mathbb{N}$, if $c(s, i) \geq 1$ at every raise and $c(\tau, s) \leq \mathrm{raises}(\tau, s)$ for every run, then $c(s, i) = 1$ at raises and 0 elsewhere. On the three states this leaves exactly c_0; c_n with $n \geq 1$ fails the ceiling on the one-step run of* smooth *from* once.

Proof. The first claim is (b)⇒(c) of Proposition 20.8. The step (once, smooth) raises nothing and costs n. □

Argued in Chapter 19 of the book.

22 Heat

Let S be finite. A *spread* is a function $p : S \to [0,1]$ with $\sum_s p(s) = 1$. Its uncertainty is $H(p) = -\sum_s p(s) \log_2 p(s)$, with $0 \log_2 0 = 0$. For $f : S \to S$, the spread after the step is $f_* p(y) = \sum_{f(x)=y} p(x)$, and the *bits removed* are $\Delta_f(p) = H(p) - H(f_* p)$. The *landing pile* of f at y is $f^{-1}(y)$, and $m_f = \max_y |f^{-1}(y)|$ is the size of the biggest one. A *known state* is a spread with $p(s_0) = 1$ for one s_0. k_B is Boltzmann's constant and $T > 0$ a temperature.

Assumption 22.1 (Landauer's principle)**.** [assumed; cited] For a chosen physical memory whose states are the states of S, symmetric (its states alike apart from their labels), prepared in a spread p, and in contact with surroundings at temperature T, a step f hands to the surroundings an average heat, over that preparation, of at least $k_B T \ln 2 \cdot \Delta_f(p)$. This is Landauer's principle in the general form used in the book; its general validity is disputed, and the book names the critics. It is not claimed of single runs.

Argued in Chapter 20 of the book.

Lemma 22.2 (The even spread holds the most)**.** [shown] *If a spread q is supported on $m \geq 1$ states, then $H(q) \leq \log_2 m$, with equality exactly for the even spread on those states.*

Proof. $\ln y \leq y - 1$, with equality only at $y = 1$. For each state with $q(x) > 0$ put $y = 1/(m\, q(x))$: $q(x) \ln \frac{1}{m q(x)} \leq \frac{1}{m} - q(x)$. Summing, $\ln 2 \,(H(q) - \log_2 m) \leq \sum_x (\frac{1}{m} - q(x)) \leq 0$, with equality only if every $q(x) = 1/m$ on all m states. □

Argued in Chapter 20 of the book.

Lemma 22.3 (Bits removed, state by state)**.** [shown] *For every f and spread p, $\Delta_f(p) = \sum_{x:\, p(x)>0} p(x) \log_2 \frac{f_* p(f(x))}{p(x)}$. Each term is at least 0, and the term of x is positive exactly when some $x' \neq x$ with $p(x') > 0$ has $f(x') = f(x)$. In particular $\Delta_f(p) \geq 0$, $\Delta_f(p) = 0$ when f is one-to-one, and the four equally likely cards sent $A, B \mapsto X$, $C \mapsto Y$, $D \mapsto Z$ lose $2 - \frac{3}{2} = \frac{1}{2}$ bit.*

Proof. $H(f_* p) = -\sum_y f_* p(y) \log_2 f_* p(y) = -\sum_x p(x) \log_2 f_* p(f(x))$; subtract from $H(p)$. Since $f_* p(f(x)) \geq p(x)$, with equality exactly when x shares its landing place with no other state of positive chance, each term is nonnegative, and positive in the stated case. □

Argued in Chapter 20 of the book.

Proposition 22.4 (A permanent raise from the even spread)**.** [shown] *Let $f : S \to S$ keep the box up ($r(s) = \text{yes} \Rightarrow r(f(s)) = \text{yes}$), let Y be the $m \geq 1$ states where the box is up, and let K be $k \geq 1$ states where it is down with $f(K) \subseteq Y$. For the even spread p on $Y \cup K$:*

(a) $\Delta_f(p) \geq \log_2 \frac{m+k}{m}$, with equality for some such f whenever m divides k;

(b) $\Delta_f(p) \geq \frac{2k}{m+k}$, with equality for some such f whenever $k \leq m$;

(c) (worst case) there are $x \neq x'$ in $Y \cup \{s\}$, for any $s \in K$, with $f(x) = f(x')$, and the even spread on $\{x, x'\}$ loses exactly one bit, whatever m.

Proof. (a) $H(p) = \log_2(m+k)$ and f_*p is supported on Y, so $H(f_*p) \leq \log_2 m$ by Lemma 22.2. (b) If $s_y = |f^{-1}(y) \cap (Y \cup K)|$, then $\Delta_f(p) = \sum_y \frac{s_y \log_2 s_y}{m+k}$. For integers $s \geq 1$, $s \log_2 s \geq 2(s-1)$ (equality at 1 and 2; for $s \geq 2$ the left side grows with slope $\log_2 s + 1/\ln 2 > 2$). At most m piles are nonempty, so $\sum_y (s_y - 1) \geq k$. Equality in (a): keep each $y \in Y$ in place and send k/m states of K to each; in (b): send the k states of K to k different states of Y and keep Y in place. (c) Proposition 20.11 gives the pair; the even spread on it has $H = 1$ before and 0 after. □

Argued in Chapter 20 of the book.

Proposition 22.5 (Any spread on the raise loses something)**.** [shown] *In the setting of Proposition 22.4, let $s \in K$, and let p be any spread with $p(s) > 0$ and $p(y) > 0$ for every $y \in Y$. Then $\Delta_f(p) > 0$.*

Proof. By Proposition 20.11, two states of $Y \cup \{s\}$, both of positive chance, land together. Apply Lemma 22.3. □

Argued in Chapter 20 of the book.

Corollary 22.6 (Heat on a permanent raise)**.** [shown] *Given Assumption 22.1: in the setting of Proposition 22.4, the average heat from the even spread on $Y \cup K$ is at least $k_B T \ln 2 \cdot \max(\log_2 \frac{m+k}{m}, \frac{2k}{m+k})$; from the even spread on the colliding pair of (c) it is at least $k_B T \ln 2$; and under the spread of Proposition 22.5 it is positive. For $m = k = 1$ the first floor is $k_B T \ln 2$, about 2.87×10^{-21} J at 300 K. For $m = 1000$, $k = 1$ it is $\frac{2 \ln 2}{1001} k_B T$, about $k_B T/722$, about 500 times less than for $m = 1$.*

Argued in Chapter 20 of the book.

Proposition 22.7 (Three roads, two strengths)**.** [shown] *For $f : S \to S$ and $c \in \mathbb{N}$:*

(a) $\max_p \Delta_f(p) = \log_2 m_f$, attained by the even spread on a biggest landing pile;

(b) (entropy pricing equals the halving rule) $\Delta_f(p) \leq c$ for every spread p if and only if $m_f \leq 2^c$;

(c) (Bennett's rule) the condition "if f makes two arrows meet, then $c \geq 1$" holds if and only if $\Delta_f(p) \leq c$ for every spread p supported on two states;

(d) Bennett's rule is weaker: for f sending eight states to one, $c = 1$ satisfies (c) and fails (b), which asks for $c \geq 3$.

Proof. (a) Write $q = f_*p$. Grouping the states of positive chance by landing place, $\Delta_f(p) = \sum_{y:\, q(y)>0} q(y)\, H(p_y)$, where p_y is p conditioned on $f^{-1}(y)$; by Lemma 22.2 each $H(p_y) \leq \log_2 |f^{-1}(y)| \leq \log_2 m_f$. The even spread on a biggest pile loses $\log_2 m_f$. (b) follows from (a), as $c \in \mathbb{N}$. (c) A spread on two states loses 1 bit if they merge with equal chances and at most 1 in any case, and loses 0 if they do not merge. (d) $m_f = 8$. □

Argued in Chapter 20 of the book.

Proposition 22.8 (No floor on a known state)**.** [shown] *For every f and every known state p, $\Delta_f(p) = 0$. Given Assumption 22.1, the bound it gives for a step from a known state is that the average heat is at least 0.*

Proof. p and f_*p are both known states, and a known state has $H = 0$. □

Argued in Chapter 20 of the book.

Assumption 22.9 (A fixed device and the second law)**.** [assumed] Let a fixed physical device send each input x of a finite set X to one output. Write $\sigma(x)$ for the average entropy it produces on input x known, and $\sigma(p)$ for that produced on a spread p of inputs. Then $\sum_x p(x)\sigma(x) = \sigma(p) + k_B \sum_x p(x) \ln \frac{1}{p(x)}$, and $\sigma(p) \geq 0$. When dissipated work is T times entropy produced, the same holds for dissipated work divided by T.

Argued in Chapter 20 of the book.

Proposition 22.10 (Some input pays)**.** [shown] *Given Assumption 22.9: if a fixed device sends K inputs to the same output, then $\sigma(x) \geq k_B \ln K$ for some one of them, and, when dissipated work is T times entropy produced, the dissipated work on that input is at least $k_B T \ln K$.*

Proof. For the even spread on the K inputs, $\frac{1}{K} \sum_x \sigma(x) = \sigma(p) + k_B \ln K \geq k_B \ln K$, and the largest term is at least the average. □

Argued in Chapter 20 of the book.

Example 22.11 (The newcomer need not collide)**.** [shown] On states $1, 2, 3, 4$ with the box down only at 1, let $f(1) = 2$, $f(2) = 3$, $f(3) = 3$, $f(4) = 4$. Then f keeps the box up, raises it at 1, and makes two arrows meet only at 3, from 2 and 3. A spread with all its chance on states where the box is down loses no bits under f.

Argued in Chapter 20 of the book.

Theorem 22.12 (Every scale pays the same way)**.** [shown] *Let $F : X' \to X'$ be a fine step, $X \subseteq X'$ the fenced fine states, $\pi : X \to C$ onto (each coarse state is a fence with at least one fine state in it), and $g : C \to C$ a coarse step that follows the fine one: $\pi(F(x)) = g(\pi(x))$ whenever $x \in X$ and $F(x) \in X$. If g makes two arrows meet, then at least one of the following holds: (a) F makes two arrows meet on X; (b) $F(x) \notin X$ for some $x \in X$; (c) some run of F from a state of X visits infinitely many distinct states.*

Proof. Suppose none holds. Then F maps X into X one-to-one and every run from $x \in X$ visits finitely many states, so $F^i(x) = F^j(x)$ for some $i < j$; cancelling F i times gives $F^{j-i}(x) = x$. So every $x \in X$ has a period $p_x \geq 1$. Let $g(c) = g(c')$ with $c \neq c'$, and pick $a \in \pi^{-1}(c)$, $b \in \pi^{-1}(c')$. With $n = p_a p_b$, $F^n(a) = a$ and $F^n(b) = b$, so $c = \pi(F^n a) = g^n(c) = g^{n-1}(g(c)) = g^{n-1}(g(c')) = g^n(c') = c'$, a contradiction. □

Argued in Chapter 20 of the book.

Assumption 22.13 (Local detailed balance for the walker)**.** [assumed; cited] A walker on $\mathbb{Z}$ steps $+1$ with chance $p \in (0, 1)$ and -1 with chance $1 - p$ at each tick. A forward step hands the surroundings heat $k_B T \ln \frac{p}{1-p}$ and a backward step takes the same amount back.

Argued in Chapter 20 of the book.

Proposition 22.14 (Going faster costs more)**.** [shown] *Given Assumption 22.13, with speed $v = 2p - 1$ and heat in units of $k_B T$:*

(a) the average heat per tick is $v \ln \frac{p}{1-p} \geq 0$, with equality exactly when $p = \frac{1}{2}$;

(b) for $0 < v < 1$, the average heat per unit of distance is $\ln \frac{1+v}{1-v} \geq 2v$;

(c) covering an average distance $N \geq 0$ at speed $0 < v < 1$ takes time N/v and heat $N \ln \frac{1+v}{1-v}$, and heat times time is at least $2N^2$.

Proof. (a) The average heat per tick is $p \ln \frac{p}{1-p} - (1-p) \ln \frac{p}{1-p} = v \ln \frac{p}{1-p}$, and v and $\ln \frac{p}{1-p}$ have the same sign, both zero only at $p = \frac{1}{2}$. (b) Divide by v and use $\frac{p}{1-p} = \frac{1+v}{1-v}$. The function $\ln \frac{1+v}{1-v} - 2v$ is 0 at $v = 0$ and has derivative $\frac{2}{1-v^2} - 2 \geq 0$. (c) Multiply: $N \ln \frac{1+v}{1-v} \cdot \frac{N}{v} \geq 2N^2$. □

Argued in Chapter 20 of the book.

23 Keeping without the heat

A run *takes fresh states forever* when it visits infinitely many distinct states.

Proposition 23.1 (A cell written once)**.** [shown] *(a) Let S be finite, $W \subseteq S$ the written states, $\mathsf{b} \notin W$ a blank state, and let a write $w : S \to S$ satisfy $w(W) \subseteq W$ and $w(\mathsf{b}) \in W$. Then two distinct states of $W \cup \{\mathsf{b}\}$ have the same image. This covers the three-state cell ($\mathsf{b} \mapsto 0$, $0 \mapsto 0$) and every cell with added remembering states. (b) On $\mathbb{Z}$, the step $n \mapsto n+1$ with the box up exactly on $\{n \geq 1\}$ is one-to-one and onto, raises the box at 0 on the first step, keeps it up, and never returns to a state it has visited.*

Proof. (a) $|W| + 1$ states land in W. (b) By inspection. □

Argued in Chapter 21 of the book.

Remark 23.2. A write defined only on b, with no next step at a written state, makes no two arrows meet. The argument in (a) uses that the write is defined on every state.

Proposition 23.3 (Noisy steps)**.** [shown] *Let $P : S \times S \to [0,1]$ with $\sum_y P(x,y) = 1$ for every x (sums over infinite sets are sums of nonnegative terms). A set $Y \subseteq S$ is* closed *(a reading that stays) when $P(x,y) = 0$ for $x \in Y$, $y \notin Y$. P* keeps the even spread *when $\sum_x P(x,y) = 1$ for every y.*

(a) If P keeps the even spread and Y is finite and closed, then $P(x,y) = 0$ for all $x \notin Y$, $y \in Y$.

(b) If $|S| = N$ is finite, and $q(y) = \frac{1}{N} \sum_x P(x,y)$ is one step from the even spread, then $\sum_y q(y)^2 \geq \frac{1}{N}$, with equality exactly when P keeps the even spread; otherwise also $H(q) < \log_2 N$. In particular, if some closed Y has $P(x,y) > 0$ for some $x \notin Y$, $y \in Y$, the step makes the even spread lumpier.

(c) If P keeps the even spread, then $\sum_y q(y)^2 \leq \sum_x p(x)^2$ for every spread p on a countable S, where $q(y) = \sum_x p(x) P(x,y)$.

(d) If π is a spread with $\sum_x \pi(x) P(x,y) = \pi(y)$ for every y and $\pi(x) > 0$ for every x, and Y is closed, then $P(x,y) = 0$ for all $x \notin Y$, $y \in Y$.

Proof. (a) $\sum_{y \in Y} \sum_x P(x,y) = |Y|$, while $\sum_{y \in Y} \sum_{x \in Y} P(x,y) = \sum_{x \in Y} 1 = |Y|$ by closedness. So $\sum_{x \notin Y} \sum_{y \in Y} P(x,y) = 0$, and each term is ≥ 0. (b) Let $c_y = \sum_x P(x,y)$, so $\sum_y c_y = N$ and $\sum_y (c_y - 1)^2 = \sum_y c_y^2 - N$. Then $\sum_y q(y)^2 = \frac{1}{N^2} \sum_y c_y^2 \geq \frac{1}{N}$, with equality exactly when

every $c_y = 1$. If not every $c_y = 1$, q is not even and Lemma 22.2 gives $H(q) < \log_2 N$. The last claim follows from (a). (c) $q(y) = \sum_x P(x,y)p(x)$ is an average of the $p(x)$ with weights $P(\cdot, y)$ summing to 1, so $q(y)^2 \le \sum_x P(x,y)p(x)^2$. Summing over y and using $\sum_y P(x,y) = 1$ gives $\sum_x p(x)^2$. (d) $\pi(Y) = \sum_{y\in Y}\sum_x \pi(x)P(x,y) = \sum_{x\in Y}\pi(x) + \sum_{x\notin Y}\pi(x)\sum_{y\in Y}P(x,y)$, and $\pi(Y) \le 1$ is finite, so the last sum is 0; with $\pi(x) > 0$, every $P(x,y) = 0$. □

Argued in Chapter 21 of the book.

Proposition 23.4 (Quantum steps)**.** [shown] *Let $\mathcal{H}$ be a finite-dimensional complex inner product space, $Y \subseteq \mathcal{H}$ a subspace (the YES directions), and $Y^\perp$ the NO directions.*

(a) If U is unitary and $UY \subseteq Y$, then $UY^\perp \subseteq Y^\perp$. A unitary step cannot raise a reading that stays.

(b) On $\ell^2(\mathbb{Z})$ with basis (e_n), the shift $Ue_n = e_{n+1}$ is unitary, maps $Y = \overline{\text{span}}\{e_n : n \ge 1\}$ into Y, and maps $e_0 \in Y^\perp$ into Y.

(c) Let Φ be a positive, trace-preserving linear map on operators of $\mathcal{H}$ with $\Phi(\mathbf{1}) = \mathbf{1}$ (it keeps the even spread), and suppose Φ maps every density operator supported in Y to one supported in Y. Then Φ maps every density operator supported in $Y^\perp$ to one supported in $Y^\perp$.

Proof. (a) U is one-to-one and $\dim UY = \dim Y$, so $UY = Y$. For $v \in Y^\perp$ and $y \in Y$, $\langle Uv, Uy\rangle = \langle v, y\rangle = 0$, so $Uv \perp UY = Y$. (b) By inspection. (c) Let P_Y and $P_\perp$ be the projections. $\Phi(P_Y)$ is supported in Y and has trace $\dim Y$, so $\text{Tr}(P_Y\Phi(P_Y)) = \dim Y = \text{Tr}(P_Y\Phi(\mathbf{1}))$, hence $\text{Tr}(P_Y\Phi(P_\perp)) = 0$. A density operator ρ supported in $Y^\perp$ has $0 \le \rho \le P_\perp$, so $0 \le P_Y\Phi(\rho)P_Y \le P_Y\Phi(P_\perp)P_Y$, which has trace 0; so $P_Y\Phi(\rho)P_Y = 0$, and since $\Phi(\rho) \ge 0$, $\Phi(\rho)P_Y = 0$. □

Argued in Chapter 21 of the book.

Theorem 23.5 (How much room keeping takes)**.** [shown] *Let $F : S \to S$ and let $Z \subseteq S$ be the states holding the fixed value, with $F(x) \in Z$ for every x. Let $s_1, \dots, s_N$ be distinct starts, L of them in Z, with $F(s) = s$ for each start $s \in Z$. Let $T \ge 0$ and suppose the promise: for all starts s, s' and $0 \le i, j < T$, $F(F^i(s)) = F(F^j(s')) \Rightarrow F^i(s) = F^j(s')$. Then*

$$|S| \;\ge\; N + T(N - L).$$

The bound is attained: for every N, $L \le N$ and T there are S, F, Z and starts meeting every condition with $|S| = N + T(N - L)$. For $L = 1$ the least number of states is $1 + (T+1)(N-1)$, and a memory of B bits needs $B \ge \log_2\big(1 + (T+1)(N-1)\big)$.

Proof. Walking back: if $F^i(s) = F^j(s')$ with $i \le j \le T$, then $s = F^{j-i}(s')$, by applying the promise i times. Consider the positions (s,k) with $0 \le k \le T$ for each start $s \notin Z$, and $(s,0)$ for each start $s \in Z$: there are $N + T(N-L)$ of them. Suppose $F^i(s) = F^j(s')$ for two positions, $i \le j$. If $s, s' \notin Z$, then $s = F^{j-i}(s')$, which lies in Z if $j > i$; so $j = i$ and $s = s'$. If $s \in Z$, then $i = 0$ and $F^j(s) = s = F^j(s')$, so walking back j steps gives $s = s'$. If $s' \in Z$, then $j = 0 = i$ and $s = s'$. So the positions give distinct states. For attainment, give each start in Z one state, fixed by F, and each other start a row of $T+1$ states, the first being the start and the others in Z, with F walking the row and sending its last state to a start in Z (or to itself if $L = 0$). Steps from states reached in fewer than T repetitions land on distinct states. □

Argued in Chapter 21 of the book.

Definition 23.6 (Two levers)**.** [chosen] A *two-lever box* is a set S with a reading $r : S \to \{0, 1, 2\}$ and maps $F, G : S \to S$ with $r(F(x)) = 0$ if $r(x) = 1$ and $r(F(x)) = r(x)$ otherwise, and $r(G(x)) = 0$ if $r(x) = 2$ and $r(G(x)) = r(x)$ otherwise, with starts z, a, b reading $0, 1, 2$. R_t is the set of states reached from a start by at most t pulls of F or G. The box *keeps states apart through horizon* T when F and G are each one-to-one on R_{T-1}. The levers *commute* when $F(G(x)) = G(F(x))$ for every $x \in S$.

Theorem 23.7 (Two levers)**.** [shown] *Let $T \geq 1$ and let a two-lever box keep states apart through horizon T.*

(a) $|S| \geq T + 3$, and some such box has exactly $T + 3$ states.

(b) If the levers commute, $G(a) = a$ and $F(b) = b$, then $|S| \geq 2T + 2$, and some such box has exactly $2T + 2$ states.

Proof. (a) Let Z_t be the states of R_t reading 0. For $1 \leq t \leq T$, F maps $Z_{t-1} \cup \{a\} \subseteq R_{t-1}$ one-to-one into Z_t, so $|Z_t| \geq |Z_{t-1}| + 1$; with $z \in Z_0$, $|Z_T| \geq T + 1$, and a, b add two. A box with $T + 3$ states: $a, b, c_1, \ldots, c_{T+1}$ with $z = c_1$, $F(a) = G(b) = c_1$, $F(b) = b$, $G(a) = a$, and $F(c_k) = G(c_k) = c_{\min(k+1,T+1)}$.

(b) Walking back as in Theorem 23.5, the states $F^k(a)$, $1 \leq k \leq T$, are distinct and read 0, and likewise $G^k(b)$. Commuting gives $G(F^k a) = F^k(Ga) = F^k a$ and $F(G^k b) = G^k b$. If $F^i a = G^j b$ with $i < T$, then F moves this state to $F^{i+1} a \neq F^i a$ and also holds it still, which is impossible; so $i = T$, and likewise $j = T$. If $F^T a \neq G^T b$, the two chains give $2T$ zero states, and a, b give $2T + 2$. If $F^T a = G^T b =: e$, the chains give $2T - 1$ states, and $F(e) = e$. Were $z = F^i a$ with $1 \leq i \leq T$, then $e = F^{T-i}(z) \in R_{T-1}$ and $F(F^{T-1} a) = e = F(e)$ with $F^{T-1} a \in R_{T-1}$, so $F^{T-1} a = e$: for $T = 1$ this makes a read 0, and for $T \geq 2$ it contradicts distinctness. So z is off the first chain, and likewise off the second, giving $2T - 1 + 1 + 2 = 2T + 2$. A box with $2T + 2$ states: $a, b, c_1, \ldots, c_T, d_1, \ldots, d_T$ with $z = d_1$; $F(a) = c_1$, $F(c_k) = c_{\min(k+1,T)}$, F fixes b and every d_k; $G(b) = d_1$, $G(d_k) = d_{k+1}$ for $k < T$, $G(d_T) = c_T$, G fixes a and every c_k. A direct check gives the reading rules, commuting, and one-to-one levers on $R_{T-1} = \{a, b, c_1, \ldots, c_{T-1}, d_1, \ldots, d_T\}$. For $T = 1$ this is the book's four-place box. □

Argued in Chapter 21 of the book.

Remark 23.8. Without the conditions $G(a) = a$ and $F(b) = b$, the book argues more briefly, from the paths the idle lever traces and the loops they fall into, that commuting levers still need $2T + 2$ states. That argument is not restated here as a result.

Proposition 23.9 (Two levers, forever)**.** [shown] *Let $T \geq 1$ and let F and G in a two-lever box be one-to-one on all of S.*

(a) If the levers commute, $G(a) = a$ and $F(b) = b$, then $|R_T| \geq 2T + 3$, and some such box has $|R_T| = 2T + 3$ at every horizon.

(b) Without commuting, some such box has $|R_T| = T + 3$, the least Theorem 23.7(a) allows.

Proof. (a) If $F^i a = F^j a$ with $i < j$, cancelling F i times gives $a = F^{j-i} a$, which reads 0: impossible. So $a, Fa, F^2 a, \ldots$ are pairwise distinct, and likewise $b, Gb, G^2 b, \ldots$. As in Theorem 23.7(b), G fixes every $F^k a$ and F fixes every $G^k b$. If $F^i a = G^j b$, then $F^{i+1} a = F(G^j b) = G^j b = F^i a$, which distinctness forbids; so the chains share no state. The states $a, b, F^k a, G^k b$ for $1 \leq k \leq T$ are $2T + 2$ distinct states of R_T. If z is on neither chain it is one more. If $z = F^k a$ with $1 \leq k \leq T$, then $F^T z = F^{T+k} a \in R_T$ lies on the first chain beyond $F^T a$,

so it is new; likewise on the second chain. So $|R_T| \geq 2T+3$. The box with states $A_0, A_1, \ldots$ and $B_0, B_1, \ldots$, the first lever sending $A_n \mapsto A_{n+1}$ and fixing every B_n, the second sending $B_n \mapsto B_{n+1}$ and fixing every A_n, $r(A_0) = 1$, $r(B_0) = 2$, $r = 0$ elsewhere, and starts $a = A_0$, $b = B_0$, $z = B_1$, commutes, is one-to-one, and has $R_T = \{A_0, \ldots, A_T, B_0, \ldots, B_{T+1}\}$, with $2T+3$ states.

(b) Take states a, b and $L_0, L_1, \ldots$ reading 1, 2 and 0, with $z = L_0$, $F(a) = L_0$, $F(b) = b$, $G(b) = L_0$, $G(a) = a$, and $F(L_n) = G(L_n) = L_{n+1}$. Both levers are one-to-one, and $R_T = \{a, b, L_0, \ldots, L_T\}$ has $T+3$ states. The levers do not commute: $F(G(a)) = L_0$ and $G(F(a)) = L_1$. □

Argued in Chapter 21 of the book.

Theorem 23.10 (The two-bit grid)**.** [shown] *Let $r : S \to \{0,1\}^2$ (each bit set or cleared), F clear the first bit and keep the second ($r(Fx) = (0, r_2(x))$), G clear the second and keep the first, with four starts, one for each reading. Let R_t be the states reached from the starts by at most t pulls, and suppose F and G are one-to-one on R_t for every $t < T$. Then*

$$|R_T| \;\geq\; 1 + \frac{(T+2)(T+3)}{2}.$$

The grid attains it for every t: the states blank *and $(i,j) \in \mathbb{N}^2$, with $F(i,j) = (i+1, j)$, $G(i,j) = (i, j+1)$, both fixing* blank*, the reading of (i,j) being whether $i = 0$ and whether $j = 0$,* blank *reading $(0,0)$, and starts* blank*, $(0,0), (1,0), (0,1)$. Its levers commute and are one-to-one everywhere, and it reaches exactly $\{\text{blank}\} \cup \{(i,j) : i + j \leq t+1\}$ in t pulls, which has $1 + \frac{(t+2)(t+3)}{2}$ states.*

Proof. Let A_t, B_t, Z_t be the states of R_t reading $(0,1)$, $(1,0)$, $(0,0)$. For $1 \leq t \leq T$, F maps A_{t-1} together with the start reading $(1,1)$ one-to-one into A_t, so $|A_t| \geq |A_{t-1}| + 1$ and $|A_t| \geq t+1$; likewise $|B_t| \geq t+1$. G maps $Z_{t-1} \cup A_{t-1}$ one-to-one into Z_t, so $|Z_t| \geq |Z_{t-1}| + t$ and $|Z_T| \geq 1 + \frac{T(T+1)}{2}$. With the start reading $(1,1)$, $|R_T| \geq 1 + 2(T+1) + 1 + \frac{T(T+1)}{2} = 1 + \frac{(T+2)(T+3)}{2}$. For the grid, each pull adds one to $i+j$, the starts have $i+j \leq 1$, and every point with $i+j \leq t+1$ is reached; the triangle has $\sum_{r=0}^{t+1}(r+1) = \frac{(t+2)(t+3)}{2}$ points. □

Argued in Chapter 21 of the book.

Proposition 23.11 (Keeping every history)**.** [shown] *For a machine (S, I, δ), let $\hat{S}$ be the pairs (s, η) with η a finite list of states, and $\hat{\delta}((s,\eta), i) = (\delta(s,i), s\eta)$. Each $\hat{\delta}(\cdot, i)$ is one-to-one, the first coordinate of $\hat{\delta}((s,\eta), i)$ is $\delta(s,i)$, and no run of $\hat{\delta}$ visits a state twice.*

Proof. From $(\delta(s,i), s\eta)$ the pair (s, η) is read off the list. The list grows by one at every step. □

Argued in Chapter 21 of the book.

Proposition 23.12 (The counter and the heat can agree)**.** [shown] *Let $E \subseteq I$ be the counted moves and $c : I \to \mathbb{N}$. There is $d : I \to [0, \infty)$ with $d(i) \geq 1$ for every $i \in E$ and $d(i) \leq c(i)$ for every $i \in I$ if and only if $c(i) \geq 1$ for every $i \in E$.*

Proof. If so, take $d = c$. Conversely $1 \leq d(i) \leq c(i)$ for $i \in E$. □

Argued in Chapter 21 of the book.

Theorem 23.13 (Receipts cover the merges a run shows)**.** [shown] *Let $f : S \to S$, $u \in S$, $L \in \mathbb{N}$, and $\kappa : S \to \mathbb{N}$. Let $D = \{f^t(u) : 0 \le t < L\}$ be the states the run leaves from, and* bill $= \sum_{t=0}^{L-1} \kappa(f^t(u))$*. Suppose that whenever $x \neq y$ in D have $f(x) = f(y)$, $\kappa(x) \ge 1$ or $\kappa(y) \ge 1$. Let M be the set of y with $f(x) = f(x') = y$ for some $x \neq x'$ in D. Then $|M| \le$* bill*.*

Proof. For each $y \in M$ choose $x_y \in D$ with $f(x_y) = y$ and $\kappa(x_y) \ge 1$. Distinct y give distinct x_y. So $|M| \le \sum_{x \in D} \kappa(x) \le$ bill. □

Argued in Chapter 21 of the book.

Example 23.14 (A parked box)**.** [shown] On $\{\text{no}, \text{yes}\}$ with $f(\text{no}) = f(\text{yes}) = \text{yes}$, $\kappa(\text{no}) = 1$, $\kappa(\text{yes}) = 0$, and $u = \text{no}$: for every $L \ge 1$ the bill is 1, $|M| = 0$ for $L = 1$ and $|M| = 1$ for $L \ge 2$, and the run ends in yes in both cases.

Argued in Chapter 21 of the book.

Proposition 23.15 (A running total in finite memory)**.** [shown] *Let S be finite, $c : I \to \mathbb{N}$, and $\mu : S \to \mathbb{N}$ with $\mu(\delta(s, i)) = \mu(s) + c(i)$ for every $s \in S$ and $i \in I$. Then $c(i) = 0$ for every i. A total kept in the machine's own finite state is honest only on states where adding the charge stays within its width.*

Proof. At a state s where μ is largest, $\mu(s) + c(i) = \mu(\delta(s, i)) \le \mu(s)$. □

Argued in Chapter 21 of the book.

Proposition 23.16 (The eight-state machine)**.** [shown] *Let $S = \mathbb{Z}/4 \times \{0, 1\}$ (four slots, each with a flag), the box being the flag, with moves* $\mathsf{stamp}(k, b) = (k+1, 1)$ *at price* 1, $\mathsf{next}(k, b) = (k+1, b)$ *at price* 0*, and* $\mathsf{jump}_j(k, b) = (j, b)$ *at price* 2 *for each j. Then S is finite, no move lowers the flag,* next *is one-to-one, the biggest landing pile has size* 2 *for* stamp *and* 4 *for each* jump_j*, and the price of each move is* $\log_2$ *of its biggest pile. With starts the states with flag* 0*, it is a Thiele machine.*

Proof. $\mathsf{stamp}^{-1}(k+1, 1) = \{(k, 0), (k, 1)\}$ and $\mathsf{jump}_j^{-1}(j, b) = \mathbb{Z}/4 \times \{b\}$. Only stamp raises the flag, and it costs 1. □

Argued in Chapter 21 of the book.

Theorem 23.17 (One move repeated)**.** [shown] *Let S be finite, $f : S \to S$, $s_0 \in S$ with the box down, and suppose the box is up at $f^n(s_0)$ for every $n \ge N$. Then there are m, n with $f^m(s_0) \neq f^n(s_0)$ and $f^{m+1}(s_0) = f^{n+1}(s_0)$. No assumption that the box stays up elsewhere is used.*

Proof. Let $O = \{f^n(s_0) : n \ge 0\}$, finite, with $f(O) \subseteq O$. If f were one-to-one on O, it would permute O, so $f^p(s_0) = s_0$ for some $p \ge 1$, and then $f^{kp}(s_0) = s_0$ with the box down for $kp \ge N$. □

Argued in Chapter 21 of the book.

Proposition 23.18 (Runs that mix moves)**.** [shown] *(a) On* $\{\text{no}, \text{yes}\}$ *with moves* swap *and* idle*, neither of which makes two arrows meet, the run* swap idle idle $\cdots$ *from* no *raises the box at its first step and keeps it up forever. (b) If the next move is a function of the state, $i = \nu(s)$, the run is one map $s \mapsto \delta(s, \nu(s))$ repeated, and Theorem 23.17 applies. (c) Let S be finite and let a*

run apply a word $w = i_1 \cdots i_\ell$ over and over. If the box is down at some state of the run and up at every state from some time on, then some move i_c makes two arrows meet at two distinct states the run visits.

Proof. (a) By inspection. (c) Rotate w so that a playing of it begins at a state where the box is down, and let g be the map of one playing. Theorem 23.17 applied to g gives two distinct visited states $x \neq y$ with $g(x) = g(y)$. Follow the two playings move by move: they start apart and end together, so at the first move where they meet, that move sends two distinct visited states to one. □

Argued in Chapter 21 of the book.

Proposition 23.19 (Two quantities that only climb)**.** [shown] *On a Thiele machine (Definition 2.2) with a run $s_0, s_1, \ldots$ along moves $i_1, i_2, \ldots$, let $\mu_n = \sum_{k \le n} c(i_k)$. Then $r(s_n) \le r(s_{n+1})$ and $\mu_n \le \mu_{n+1}$ for every n, and $\mu_{n+1} \ge \mu_n + 1$ whenever $r(s_{n+1}) \not\le r(s_n)$. If moreover c follows the halving rule ($m_{\delta(\cdot,i)} \le 2^{c(i)}$ for every i) on a finite S, then for every spread p and every list of moves τ, the bits removed by the run are at most its price: $H(p) - H(\mathrm{run}(\tau, \cdot)_* p) \le \sum_{i \in \tau} c(i)$.*

Proof. The first part is clauses (i) and (iii) of Definition 2.2 with prices in $\mathbb{N}$. For the second, the bits removed by a run are the sum of the bits removed by its steps, each on the spread it receives, and each is at most $\log_2 m_{\delta(\cdot,i)} \le c(i)$ by Proposition 22.7. □

Argued in Chapter 21 of the book.

Thesis 23.20 (The second law as the toll)**.** [thesis, not shown] The second law of thermodynamics, read as a statement about what a coarse description can keep, is the toll seen from the physics side.

Argued in Chapter 21 of the book.

Remark 23.21. [suspicion, kept apart] The book records a suspicion, not shown and used by no argument: that the event it prices on a machine is the kind of event physics calls a measurement.

Remark 23.22. [not claimed] The counter is not claimed to be heat in joules. Proposition 23.12 shows only that the two assumptions it names can hold together.

24 What's shown, what's chosen, what's argued

Proposition 24.1 (A clock passes the floor and fails the ceiling)**.** [shown] *(a) On any machine, the price list $c \equiv 1$ obeys the toll for every reading. (b) Let a machine carry a box and a price list $c : I \to \mathbb{N}$, and suppose some step from s by i raises the box. If c obeys the toll, then the run ii from s costs at least 2 and raises the box once, so $c(\tau, s) \le \mathrm{raises}(\tau, s)$ fails.*

Proof. (a) Every move costs 1. (b) $c(i) \ge 1$ by the toll; from $\delta(s, i)$ the box is up, so the second step raises nothing and costs $c(i)$. □

Argued in Chapter 22 of the book.

Proposition 24.2 (Two machines side by side)**.** [shown] *Let (S_A, I_A, δ_A) and (S_B, I_B, δ_B) carry boxes r_A, r_B that never fall and price lists c_A, c_B that obey the toll. The pair has states $S_A \times S_B$, moves $I_A \sqcup I_B$, each move acting on its own coordinate, and price c_A or c_B according to the move.*

(a) The readings $r_A \vee r_B$ (either box up) and $r_A \wedge r_B$ (both up) never fall, and the pair's price list obeys the toll for each.

(b) A run from a state with both boxes down to a state with both up costs at least 2.

(c) The reading $r_A \wedge \neg r_B$ (the first up and the second down) falls whenever the first box is up and the second machine raises its box. With such a step, it violates clause (i) of Definition 2.2.

Proof. (a) A move changes one coordinate, whose box never falls, so neither reading falls. If either reading is raised by a move, the box of the machine that move belongs to is raised, and that machine's toll charges at least 1. (b) Each box goes from down to up along the run, so some move of each machine raises its box, each costing at least 1. (c) At (s, t) with $r_A(s) =$ yes, a raising step of B at t takes the reading from yes to no. □

Argued in Chapter 22 of the book.

Proposition 24.3 (Branching steps on a finite yes side)**.** [shown] *Let $\to$ be a relation on S (a state may have several next states). Let Y be the finite set of states where the box is up, with every $y \in Y$ having at least one next state and every next state of $y \in Y$ in Y, and let some $s \notin Y$ have a next state in Y. Then two distinct states of $Y \cup \{s\}$ have a common next state.*

Proof. Choose one next state in Y for each of the $|Y| + 1$ states of $Y \cup \{s\}$; two choices coincide. □

Argued in Chapter 22 of the book.

Theorem 24.4 (A branching run after the raise)**.** [shown] *Let $x_0, x_1, \ldots$ be a run of a relation $\to$ on any set S ($x_n \to x_{n+1}$ for every n), and let $k \geq 1$ with the box down at x_{k-1} and up at every x_n, $n \geq k$. If $x_i = x_j$ for some $k \leq i < j$, then there are p, q with $x_p \neq x_q$ and $x_{p+1} = x_{q+1}$. So after the raise, the run makes two arrows meet among its own states or never repeats a state.*

Proof. Compare x_{i-d} and x_{j-d} for $d = 0, 1, \ldots, i-k+1$. They are equal at $d = 0$. At $d = i-k+1$, x_{k-1} has the box down and $x_{j-i+k-1}$, with $j - i + k - 1 \geq k$, has it up, so they differ. At the least d where they differ, $p = i - d$ and $q = j - d$ give $x_p \neq x_q$ and $x_{p+1} = x_{q+1}$. □

Argued in Chapter 22 of the book.

Remark 24.5. For branching steps Proposition 24.3 says little: in the lambda calculus every term N is one step from $(\lambda x.x)N$ and from $(\lambda y.N)M$, for any M and y not free in N. The single-run statement, Theorem 24.4, applies to one run of any kind of step, including noisy and quantum steps with infinitely many yes states. Steps in continuous time are not covered.

Remark 24.6. [not claimed] Nothing here is claimed to settle P versus NP, or any question about what computers can do in a given time.

Thesis 24.7 (The Thiele thesis)**.** [thesis] Every physical process that computes with final answers has an exact description as a machine with one next state for every state, given by a program-computable step on a countable set of states, with inputs placed by a computable map, on which the answer is a computable box that no run from any input ever turns from up to down, on accepted and rejected inputs alike.

Argued in Chapter 22 of the book.

Proposition 24.8 (What a final answer accepts)**.** [shown] *Let $f : S \to S$ on a countable S coded in $\mathbb{N}$, $\iota : \mathbb{N} \to S$, and a box r, all computable, with $r(f^n(\iota(x))) = \text{yes} \Rightarrow r(f^{n+1}(\iota(x))) = \text{yes}$ for every x, n. Let $A = \{x : \exists n\ r(f^n(\iota(x))) = \text{yes}\}$.*

(a) A is listable: some program lists exactly the elements of A.

(b) Every listable set is the set A of such a description.

(c) Given Thesis 24.7, every set that a physical process accepts with final answers is listable.

(d) The condition on rejected inputs is needed: the reading that starts at no, is yes while program x is running, and falls to no for good when it stops, is yes for good exactly on the programs that never stop, and that set is not listable.

Proof. (a) At round m, run each $x < m$ for m steps and list x if its box is up. If $x \in A$ with the box up at step n, it is up at every later step, so x is listed at round $\max(n, x) + 1$; only elements of A are listed. (b) Given a listing program, let the state hold an input and a search of the list for it; on finding it the machine steps from its stopped state to itself forever, and the box reads whether it has stopped, which nothing turns off. (c) follows from (a). (d) If a program listed the programs that never stop, running it beside the search for a stop would decide for every program whether it stops, which no program does. □

Argued in Chapter 22 of the book.

The blindfold

Proposition 24.9 (Turing's model makes no mark hold)**.** [shown] *For finite sets Q of states and Σ of symbols, a table is any map $t : Q \times \Sigma \to Q \times \Sigma \times \{\text{L}, \text{R}\}$. For every table t, every (q, a) and every $w \in \Sigma$, the map equal to t except that at (q, a) it writes w is a table. So for every mark, the model contains tables that rub it out and tables that print it with no check before; no condition of the model makes a mark, once written, stay.*

Proof. Every map of that type is a table. □

Argued in Chapter 22 of the book.

Proposition 24.10 (A machine that stops, read through whether it has stopped)**.** [shown] *Let $g : S \rightharpoonup S$ be a partial step, H the states with no next step, $\bar{g}(s) = g(s)$ for $s \notin H$ and $\bar{g}(s) = s$ for $s \in H$, and $r(s) = \text{yes}$ exactly when $s \in H$. If some start $s_0 \notin H$ reaches H, then $\bar{g}$ with one move of price 1, the box r and that start is a Thiele machine. Every stopping step, from $s \notin H$ to $g(s) \in H$, makes two arrows meet: $\bar{g}(s) = \bar{g}(g(s))$.*

Proof. A stopped state stays stopped, so the box never falls; the run from s_0 raises it; the one move costs 1. □

Argued in Chapter 22 of the book.

Theorem 24.11 (The blindfold)**.** [shown] *Let (S, I, δ) be a machine with $s_0 \in S$ and a list of moves τ such that $x = \text{run}(\tau, s_0) \neq s_0$. Let $\hat{S} = S \times \{\text{no}, \text{yes}\}$, $\hat{\delta}((s, \beta), i) = (\delta(s, i), \beta')$ with $\beta' = \text{yes}$ when $\beta = \text{yes}$ or $\delta(s, i) = x$ and $\beta' = \text{no}$ otherwise, $\hat{c} \equiv 1$, the box $\hat{r}(s, \beta) = \beta$, and starts $\{(s, \text{no}) : s \in S\}$. Then:*

(a) $(\hat{S}, I, \hat{\delta}, \hat{r}, \hat{c})$ *with those starts is a Thiele machine;*

(b) through the window $\pi(s, \beta) = s$*, every run of it is a run of* (S, I, δ)*:* $\pi(\mathrm{run}(\tau', (s, \beta))) = \mathrm{run}(\tau', s)$ *for every* τ'*,* s*,* β*;*

(c) (s, no) *and* (s, yes) *have the same picture and different readings, so no function of the picture gives the reading.*

So (S, I, δ) *is what remains of a Thiele machine when its box is covered by the window* π *and the price of every move, settling included, is set to* 0*.*

Proof. (a) β never goes from yes to no; from (s_0, no) the run τ is nonempty and its last step lands on x, so the box is up at the end; every move costs 1. (b) By induction on τ'. (c) By inspection. □

Argued in Chapter 22 of the book.